

AI for accelerated materials science

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WHO AM I?

Who am I, and why this matters to me

KING'S
College
LONDON

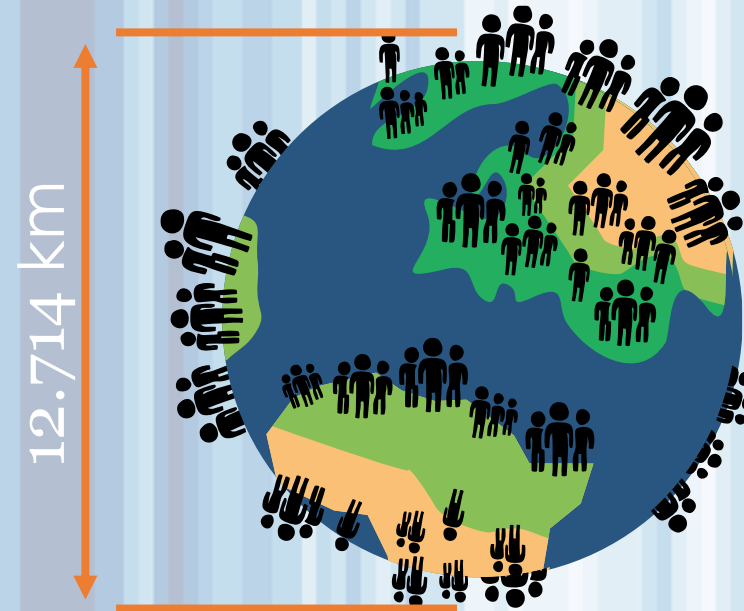


 **Swansea**
University
Prifysgol
Abertawe

A scientist using AI as a tool vs. AI doing science autonomously, where is the line?

WHY THIS MATTERS TO ME?

Global temperature change (1850-2022)



1860

1890

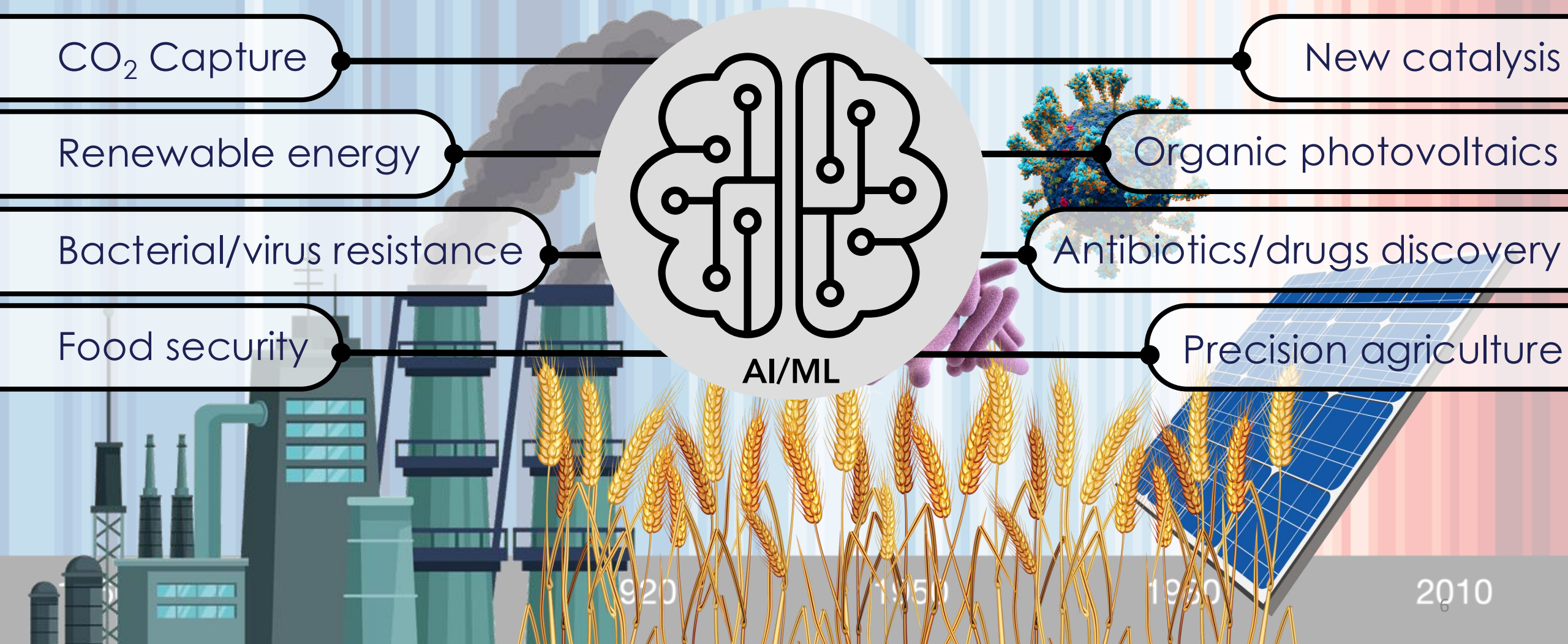
1920

1950

1980

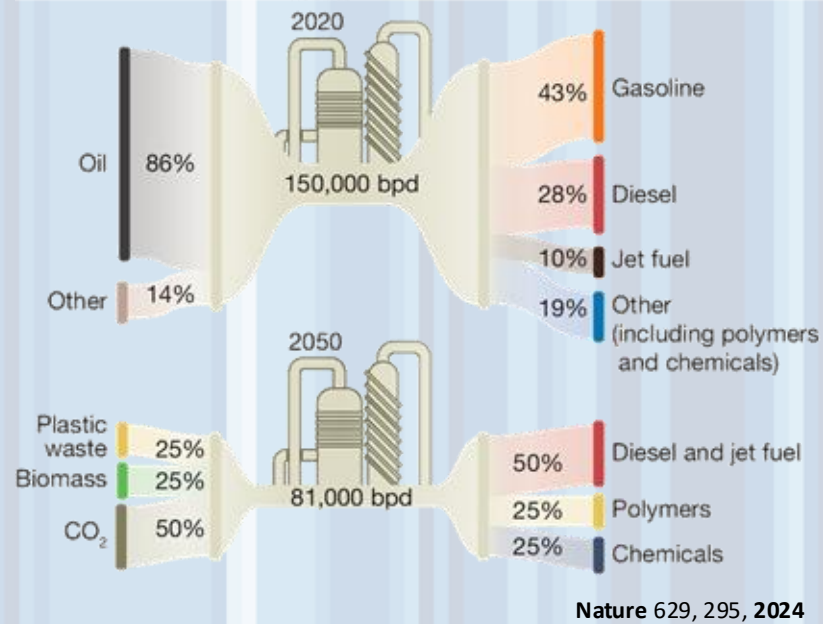
2010

Global temperature change (1850-2022)

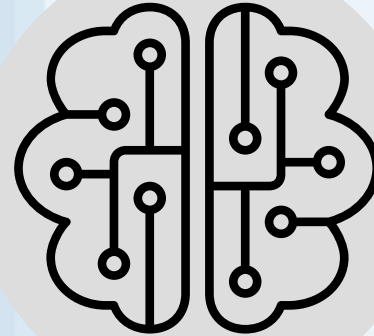


New mining of materials from waste

(Bio)refinery 2050

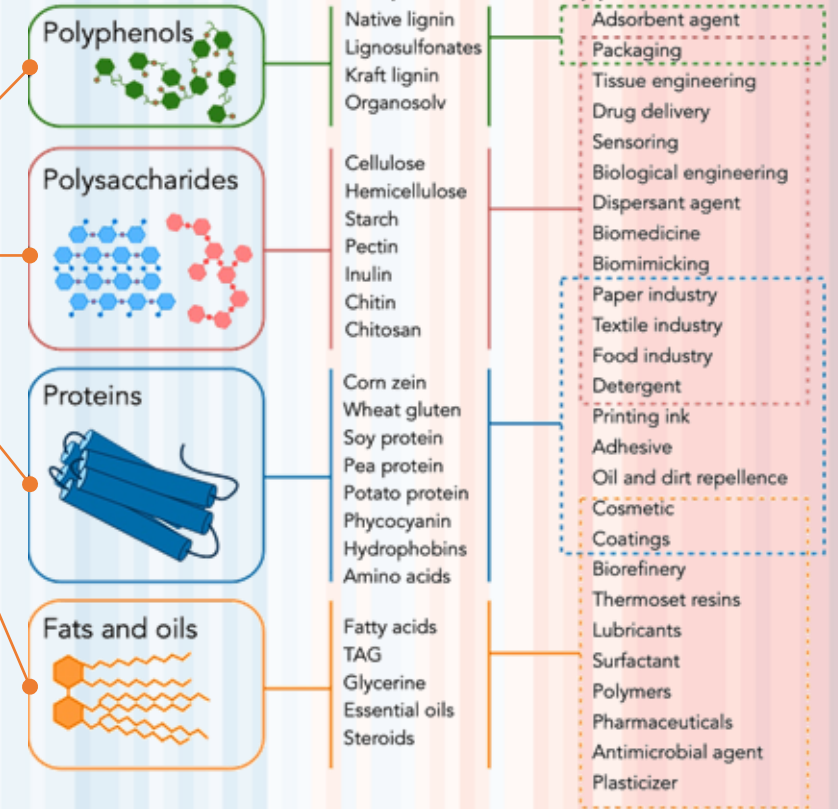


Biomass waste
140 Gt/yr



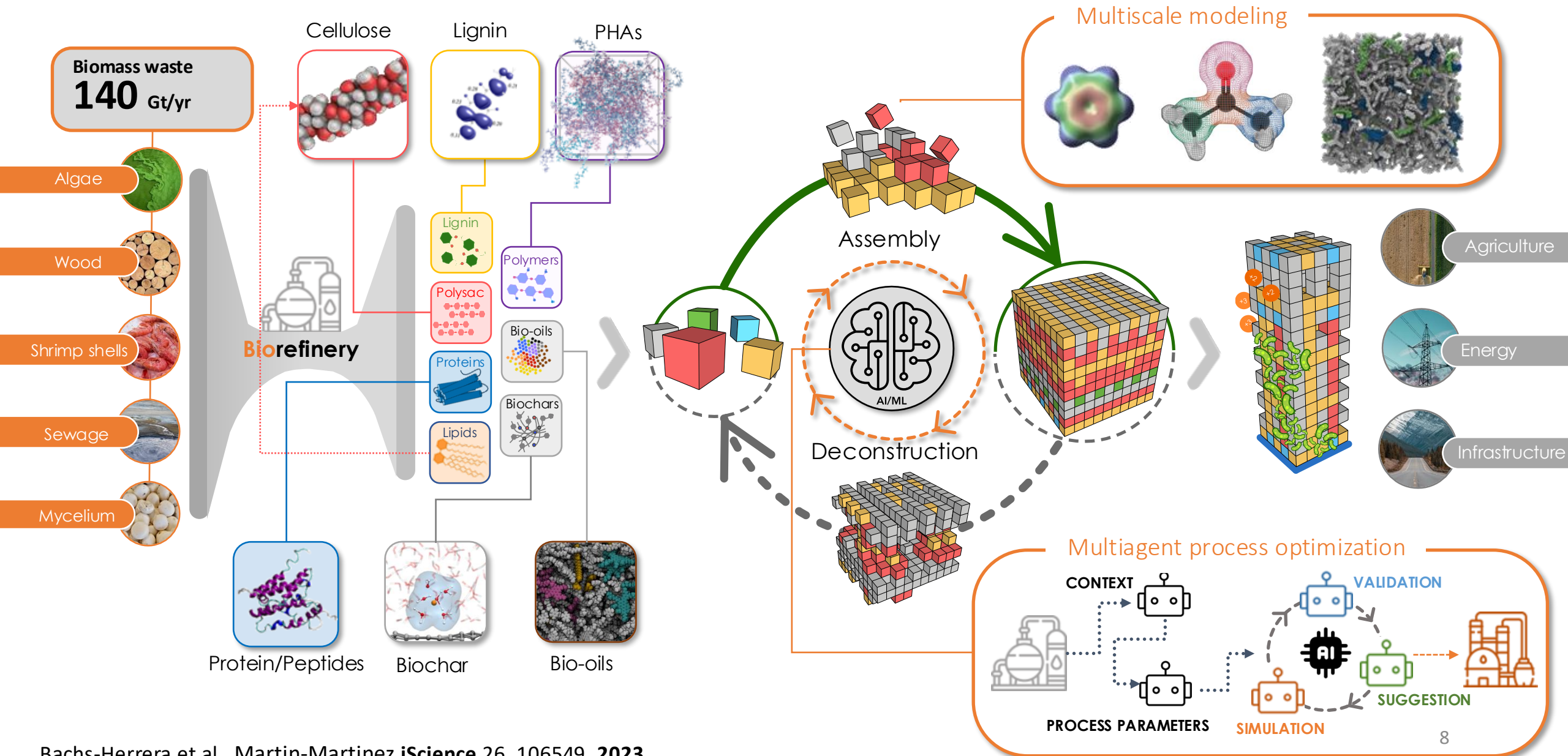
AI/ML

Waste valorization into biobased molecules and materials



Bachs-Herrera, ..., Martin-Martinez *iScience* 26, 106549, 2023

Biobased circular materials ➡ Towards an AI-driven quantum-informed biorefinery



WHAT WILL WE COVER TODAY?

WHAT WE'LL COVER

01

Why materials are a climate bottleneck

02

The traditional (still useful) paradigm of materials modeling and simulations

03

A new paradigm in materials discovery

04

The AI tool kit

05

Talking Chemistry to computers

04

Responsible AI

05

Explainable AI in Chemistry

Live poll – Use the chat

Which materials challenge do you think is most urgent for climate?

No wrong answer — this warms up the room

THE MATERIALS BOTTLENECK

Where AI plugs into the materials pipeline



Materials are a climate bottleneck



Photovoltaics

Efficiency & stability limited by materials chemistry



Batteries

Energy density, lifetime, critical-mineral dependence



Alternative fuels

Catalysts & carriers for green H₂, e-fuels



CO₂ sorbents

Capture materials that are cheap, durable, scalable



Others: sustainable infrastructure, food security, drug discovery, antibiotics ...

The problem: the discovery cycle is slow and expensive.

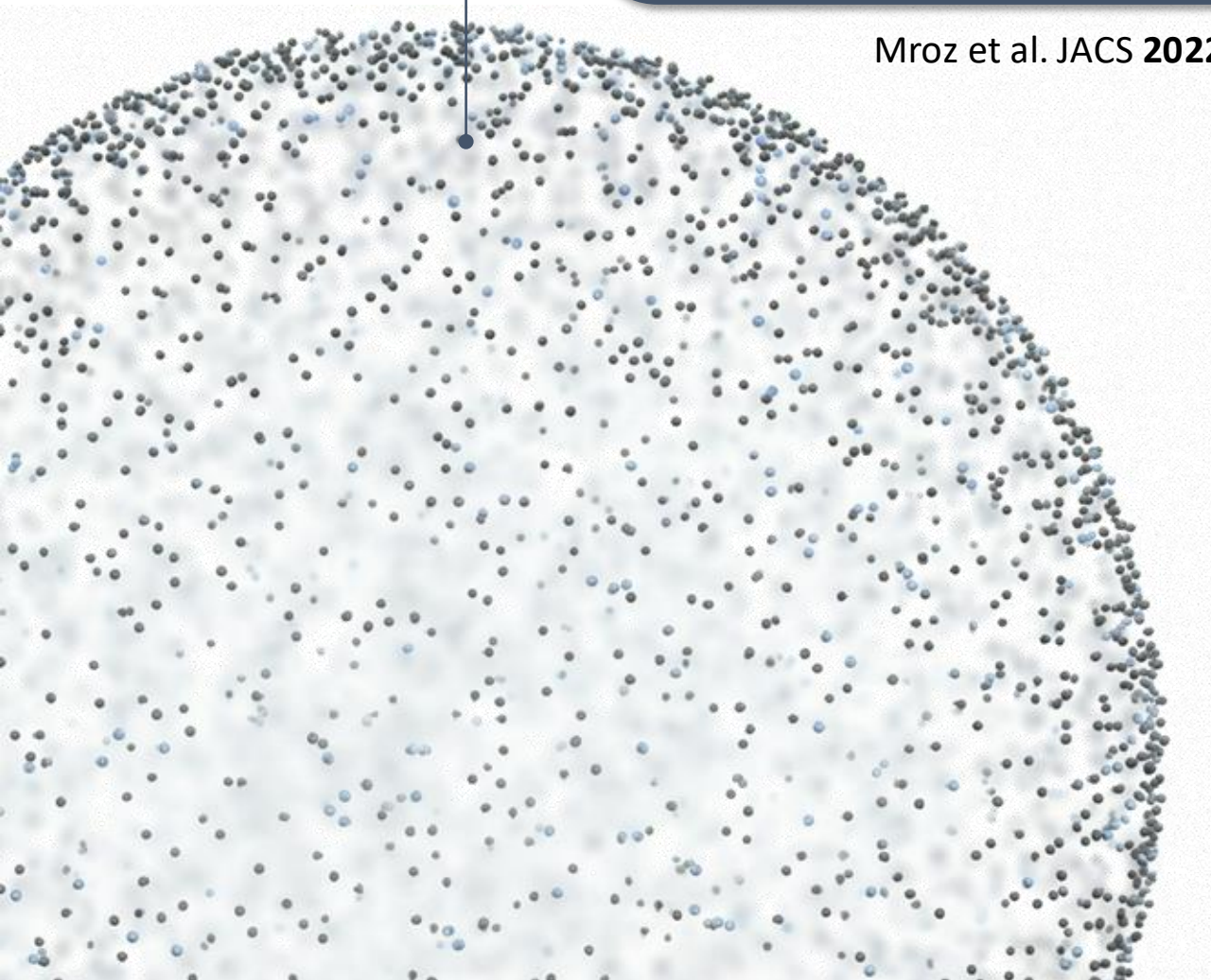
Design → synthesize → characterize → test can take years per candidate.

Mitigation AND adaptation both depend on getting the right materials, faster and more cheaply.

The materials-discovery bottleneck

10^{60} Small (drug-like) molecules for materials design

Mroz et al. JACS 2022



Combinatorial chemical space is vast
VS
Experimental validation is slow and expensive

CURIOSITY

There are more possible small organic molecules than atoms in the observable universe.

Quick quiz – True/False

1 There are more possible small organic molecules than atoms in the observable universe.

☒ True☐ False

2 A new material usually reaches the market in a few weeks.

☐ True☒ False

3 AI plugs into the pipeline at represent, predict, generate, synthesize and characterize.

☒ True☐ False

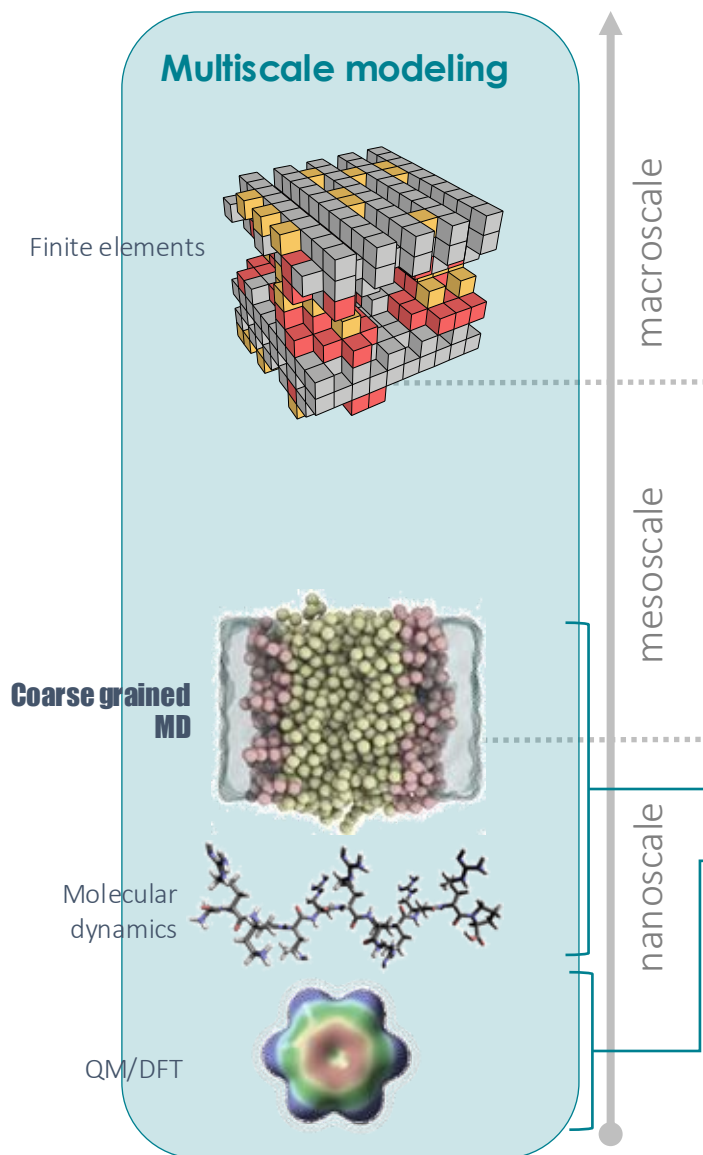
Take-home messages

WHY IT MATTERS TO CLIMATE

- 1 Materials are a climate bottleneck: mitigation and adaptation both need better materials, faster.
- 2 Chemical space is unsearchable by brute force. The use of AI reframes it as guided search.
- 3 Climate targets are on a fixed clock. Faster discovery means faster deployment.

MATERIALS MODELING AND SIMULATIONS

Modeling across scales



Molecular dynamics simulations



"All model are wrong, but some are useful."

– George E. P. Box



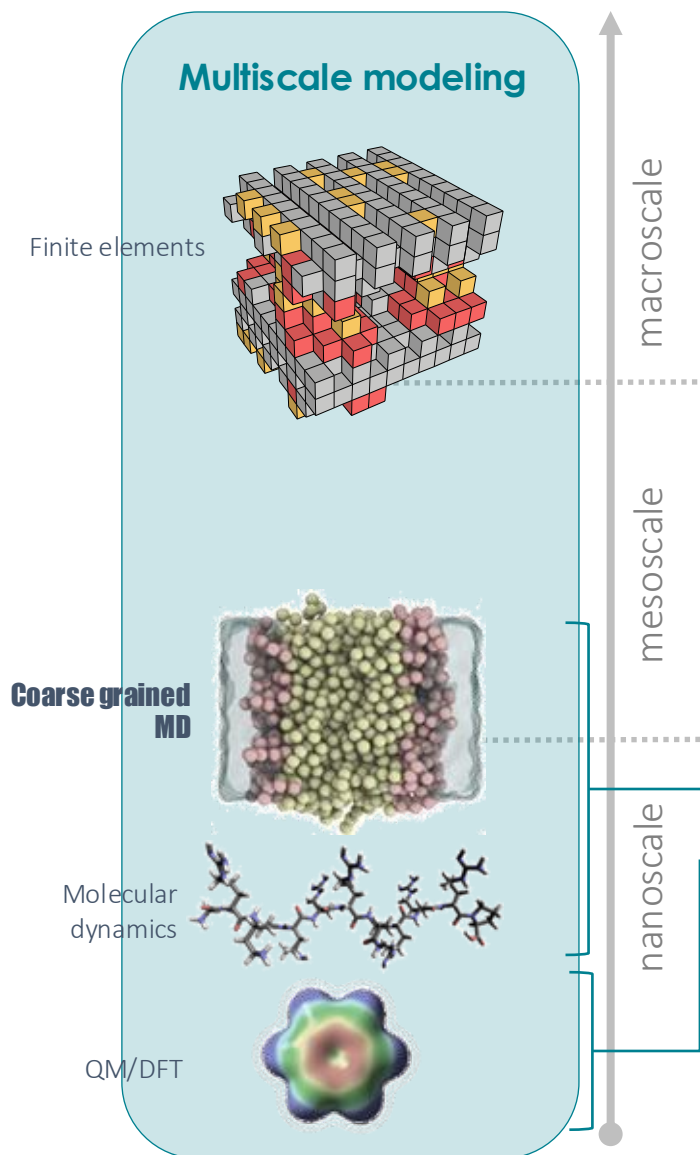
Density functional theory



"I think I can safely say that nobody understands quantum mechanics."

– Richard Feynmann

Modeling across scales



Molecular dynamics simulations



"All model are wrong, but some are useful."

– George E. P. Box

Density functional theory (DFT) – Quantum mechanics

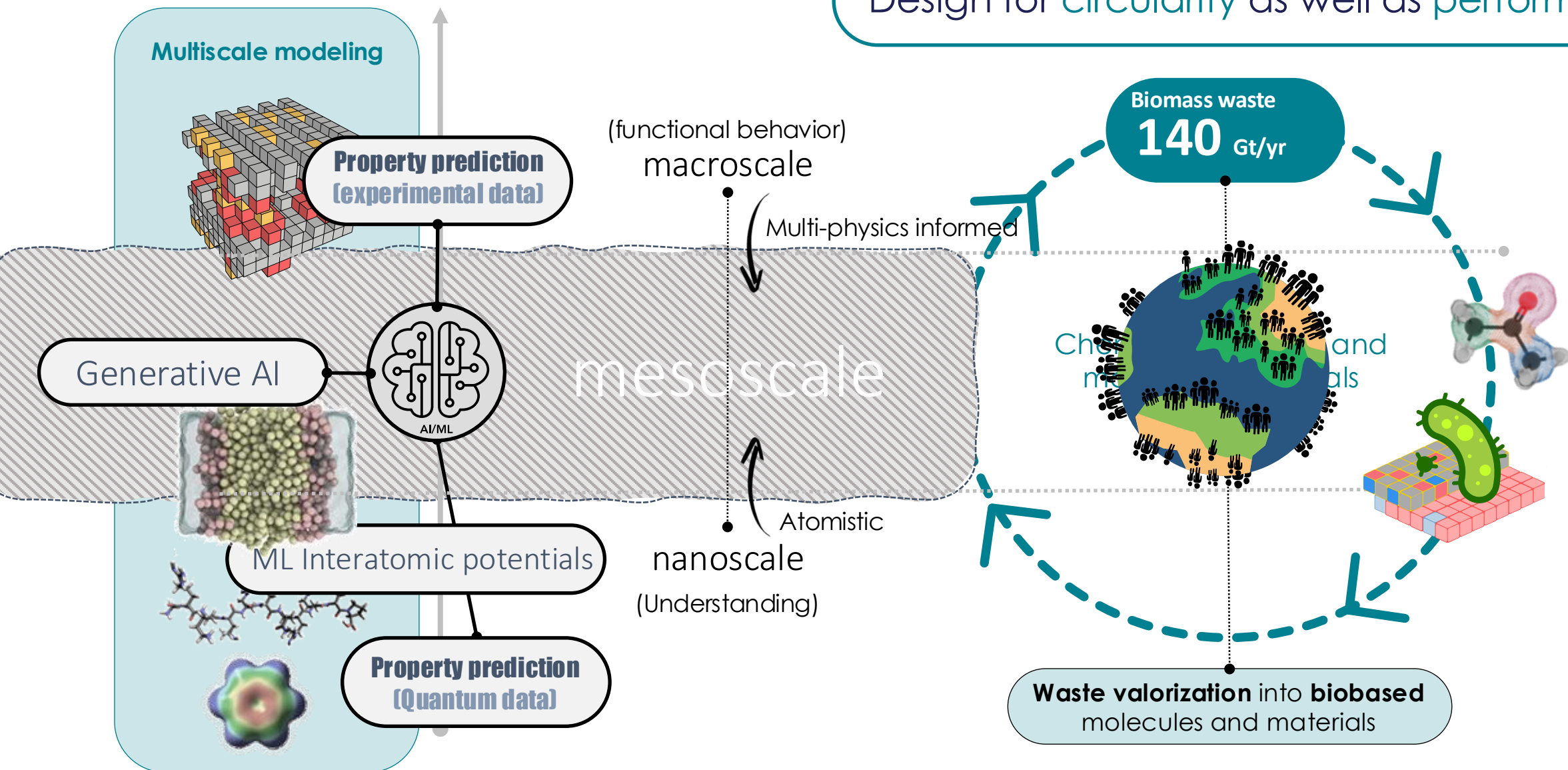


"I think I can safely say that nobody understands quantum mechanics."

– Richard Feynmann

Modeling across scales

Design for circularity as well as performance



Software for DFT, MD, and ML for Chemistry

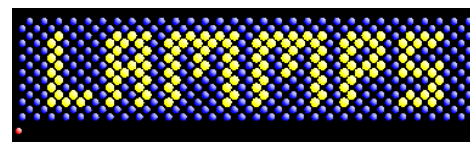
Molecular DFT



Solid state DFT



MD (All atoms and coarse grained)



Integration and cheminformatics



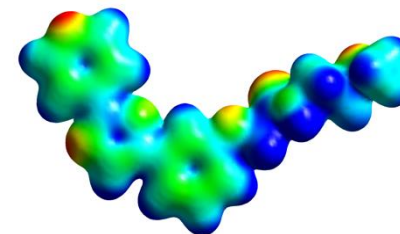
Open-Source Cheminformatics
and Machine Learning

+ atomic
simulation
environment (ASE)

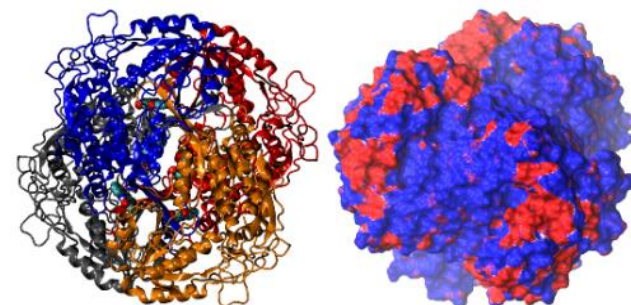
Databases to facilitate discovery in chemistry

Name	Description	Size (ca.) ^[a]	Availability
<i>Chemical structures:</i>			
ZINC ^[43, 44]	commercially available compounds	35 M	open
ChemSpider ^[45]	structures and miscellaneous data	67 M	API
SureChEMBL ^[46]	structures from ChEMBL	1.9 M	open
Super Natural II ^[47]	natural product structures	325 k	open
SAVI ^[48]	enumerated synthetically accessible structures and their building blocks	283 M	open
<i>Molecular properties:</i>			
QM7/QM7b ^[84, 85]	electronic properties (DFT)	7200	open
QM9 ^[86]	electronic properties (DFT)	134 k	open
QM8 ^[87]	spectra and excited-state properties	22 k	open
FreeSolv ^[88]	aqueous solvation energies	642	open
NIST Chemistry WebBook ^[89]	miscellaneous molecular properties	varies	open
<i>Biological data:</i>			
PubChem ^[36]	compounds and properties, emphasis on bioassay results	96 M	open
ChEMBL ^[58, 59]	compounds and bioactivity measurements	1.9 M	open
ChEBI ^[60]	compounds and biological relevance	56 k	open
PDB ^[61]	biological macromolecular structures	150 k	open
PDBBind ^[62]	protein binding affinity	20 k	open
ProTherm ^[63]	thermodynamic data for proteins	10 k	open
LINCS ^[64]	cellular interactions and perturbations	varies	open
SKEMPI ^[65]	energetics of mutant protein interactions	7 k	open
xMoDEL ^[66]	MD trajectories of proteins	1700	open
GenBank ^[67]	species' nucleotide sequences	400 k	open
DrugBank ^[68]	drug compounds, associated chemical properties, and pharmacological information	13 k	open
BindingDB ^[69]	compounds and binding measurements	750 k; 1.7 M	open
CDD ^[70]	collaborative drug discovery database for neglected tropical diseases	> 100 datasets	registration
ToxCast ^[71, 72]	compounds and cellular responses	> 4500	open
Tox21 ^[73, 74]	compounds and multiple bioassays	14 k	open

Density Functional Theory (DFT)



Molecular dynamics (MD)



SECTION CLOSER · QUIZ

Quick quiz – multiple choice

Q1

Which method is most accurate but most expensive?

A) Classical MD

C) DFT (quantum)

B) Coarse-grained MD

D) Finite elements

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Q2

Why do we use multiscale modeling?

A) One scale is enough

C) To save on citations

B) Different phenomena live at different scales

D) It is legally required

SECTION CLOSER · QUIZ

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Q3

Vs all-atom MD, coarse-grained MD reaches...

A) Smaller systems

C) Higher quantum accuracy

B) Larger systems & longer times

D) No difference

SECTION CLOSER · QUIZ

Quick quiz – multiple choice

- Q1** Which method is most accurate but most expensive?
- A) Classical MD
 - C) DFT (quantum)**
 - B) Coarse-grained MD
 - D) Finite elements
- Q2** Why do we use multiscale modeling?
- A) One scale is enough
 - C) To save on citations
 - B) Different phenomena live at different scales**
 - D) It is legally required
- Q3** Vs all-atom MD, coarse-grained MD reaches...
- A) Smaller systems
 - B) Larger systems & longer times**
 - C) Higher quantum accuracy
 - D) No difference

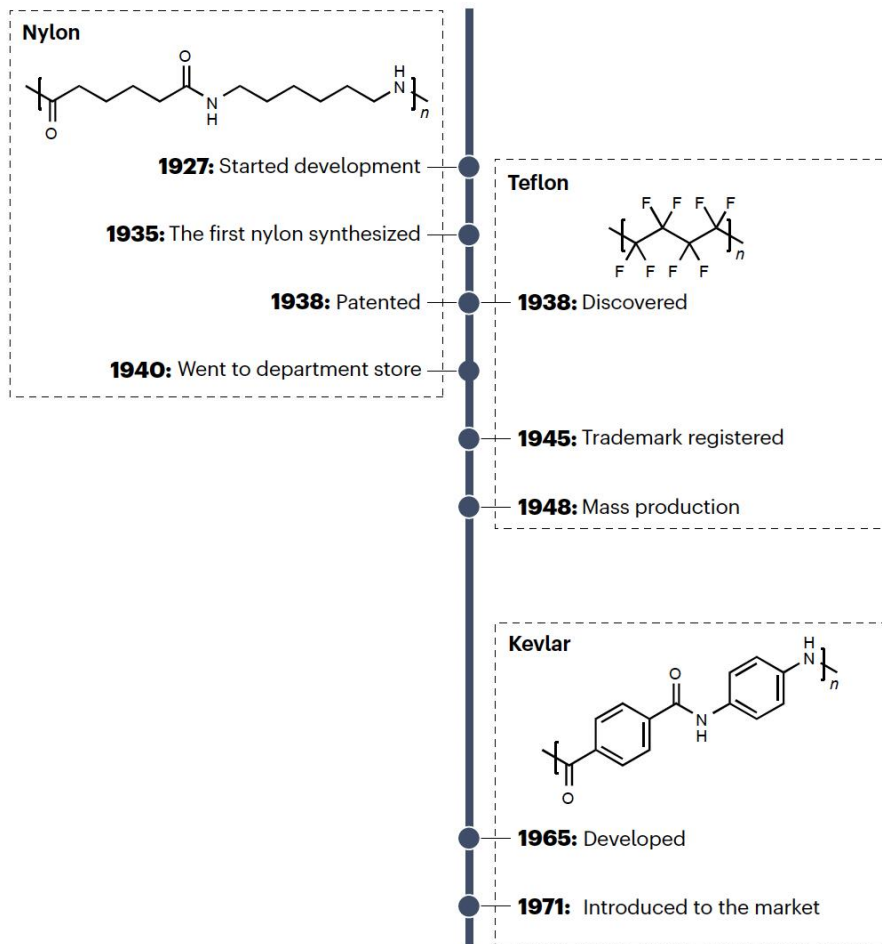
Take-home messages - The cheapest place to fail is on a computer.

WHY IT MATTERS TO CLIMATE

- 1 No single scale suffices, and multiscale modeling is needed.
- 2 We must predict degradation and recyclability: circularity, not just performance.
- 3 Data is the substrate AI builds on. Good data makes good models.

A NEW PARADIGM IN MOLECULAR AND MATERIALS DISCOVERY

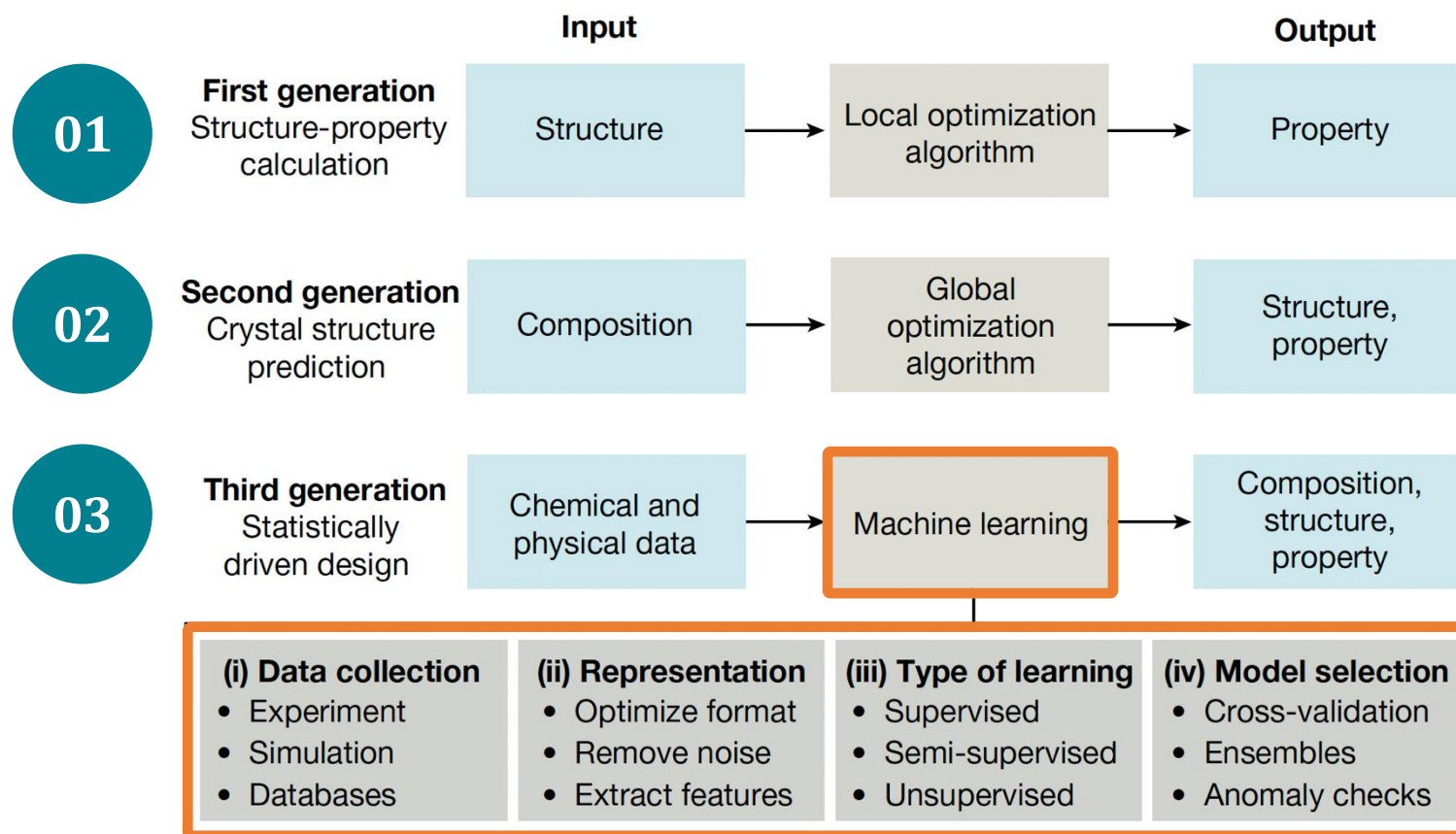
Polymer innovations over the past two centuries.



-
- A vertical timeline with a dark blue line and circular markers. The milestones are listed to the right of the line, with the year in bold and the event in regular text. The timeline starts at the top with 1833 and ends at the bottom with 2020.
- 1833:** Berzelius coins the term 'polymer'
- 1844:** Goodyear patents the vulcanization of rubber
- 1846:** Schonbein synthesizes nitrocellulose
- 1856:** Parkes patents the first thermoplastic
- 1892:** Cross, Bevan and Beadle patent 'viscose'
- 1907:** Baekeland creates the first synthetic plastic
- 1912:** Klatte invents the production process of PVC
- 1920:** Staudinger publishes 'uber polymerization'
- 1933:** Fawcett and Gibson discover LDPE
- 1937:** Bayer discovers polyurethane
- 1938:** Carothers patents nylon, i.e., Polyamide 6.6, and Plunkett discovers Teflonane
- 1944:** McIntire discovers polystyrene
- 1954:** Ziegler-Natta catalyst discovered for polypropylene production
- 1965:** Kwolek creates Kevlar
- 1977:** Heeger, MacDiarmid and Shirakawa develop polyacetylene, the first conductive polymer
- 2011:** Launch of Materials Genome Initiative
- 2020:** 100-year anniversary of modern polymer science

From trial-and-error to data-driven design:

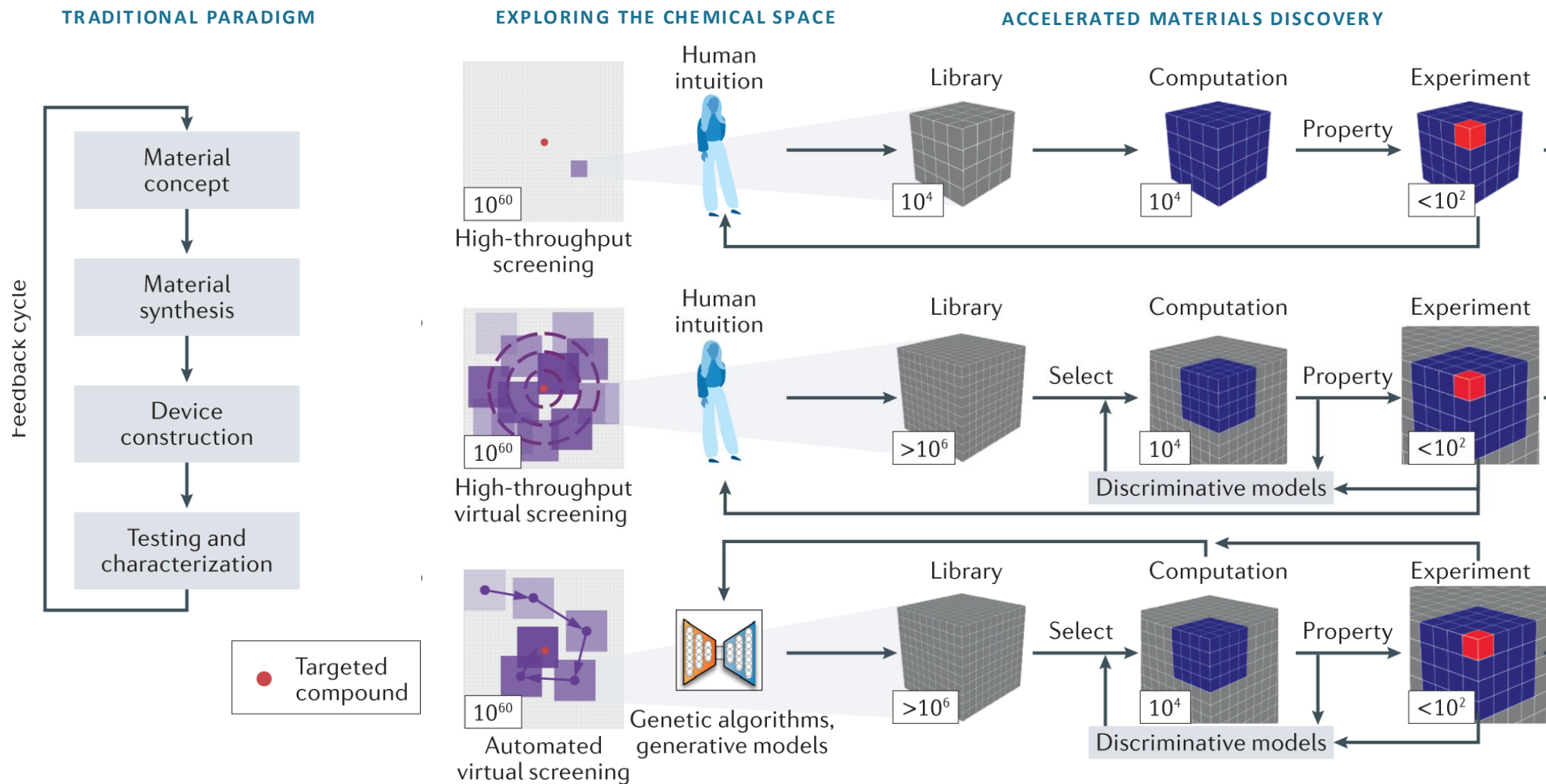
evolution of the research workflow in computational chemistry



Butler, Davies, Cartwright, Isayev, Walsh, "Machine learning for molecular and materials science," Nature 559, 547–555 (2018). DOI: 10.1038/s41586-018-0337-2

New paradigm in materials discovery

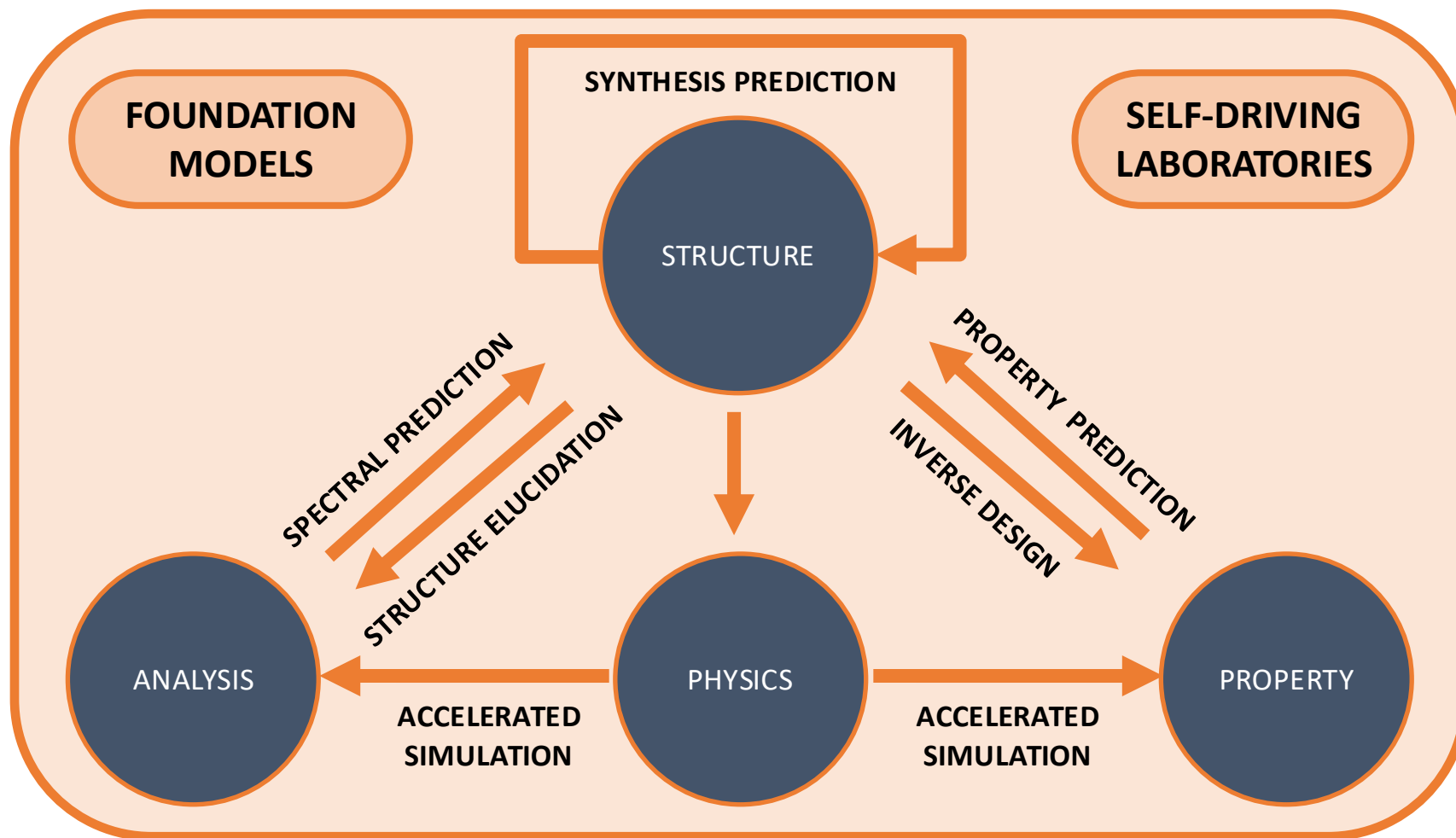
A NEW PARADIGM



Yao, Lum, Johnston, Mejia-Mendoza, Zhou, Wen, Aspuru-Guzik, Sargent, Seh, "Machine learning for a sustainable energy future," Nat. Rev. Mater. 8, 202–215 (2023).

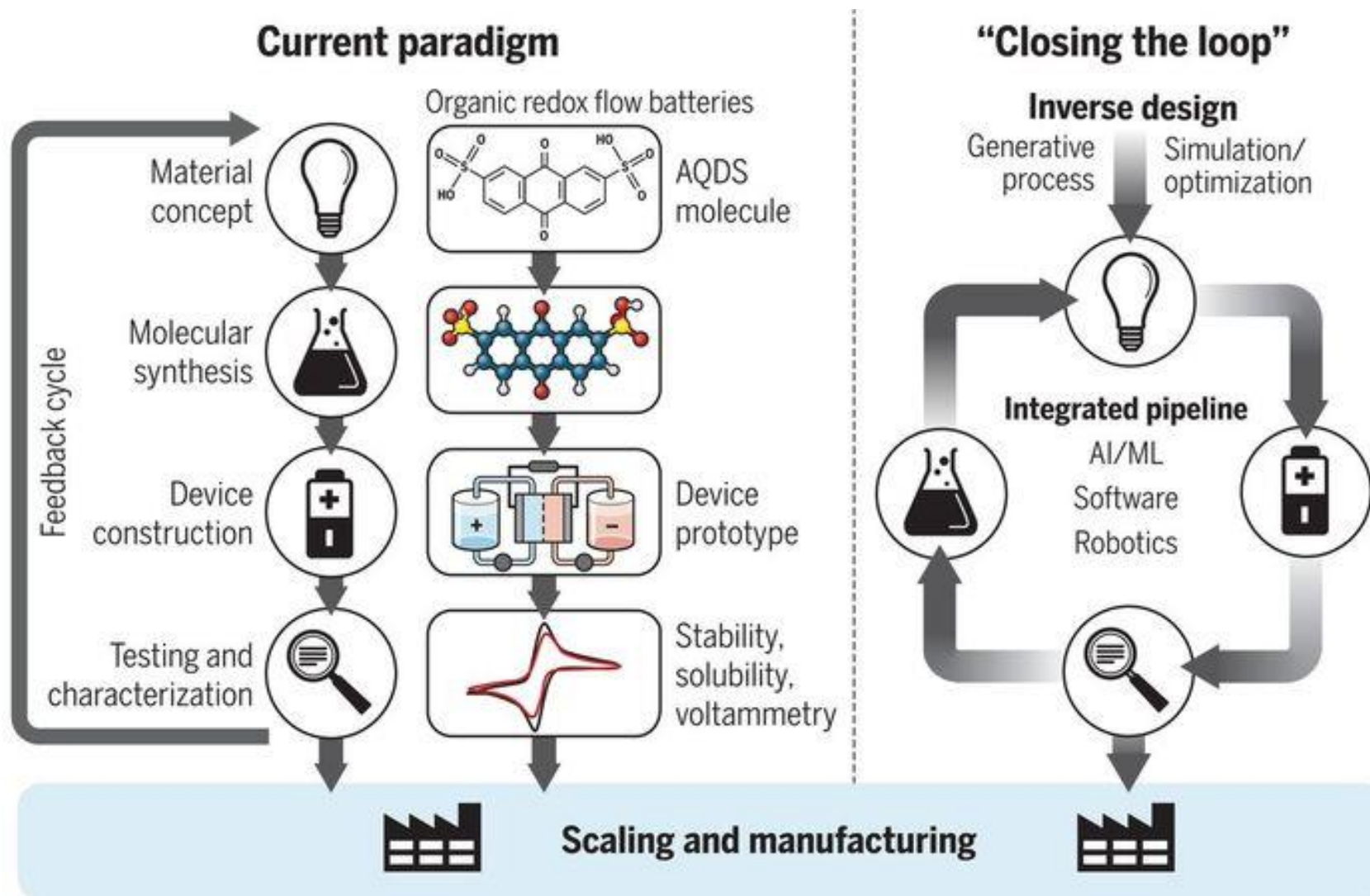
Machine learning for chemistry and materials

A NEW PARADIGM

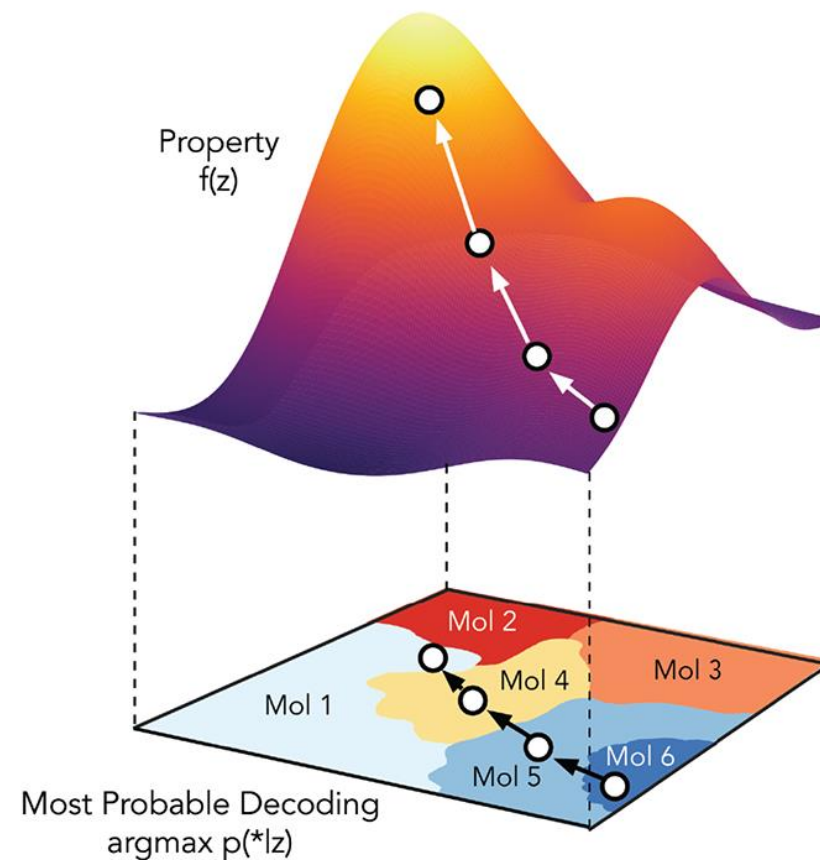
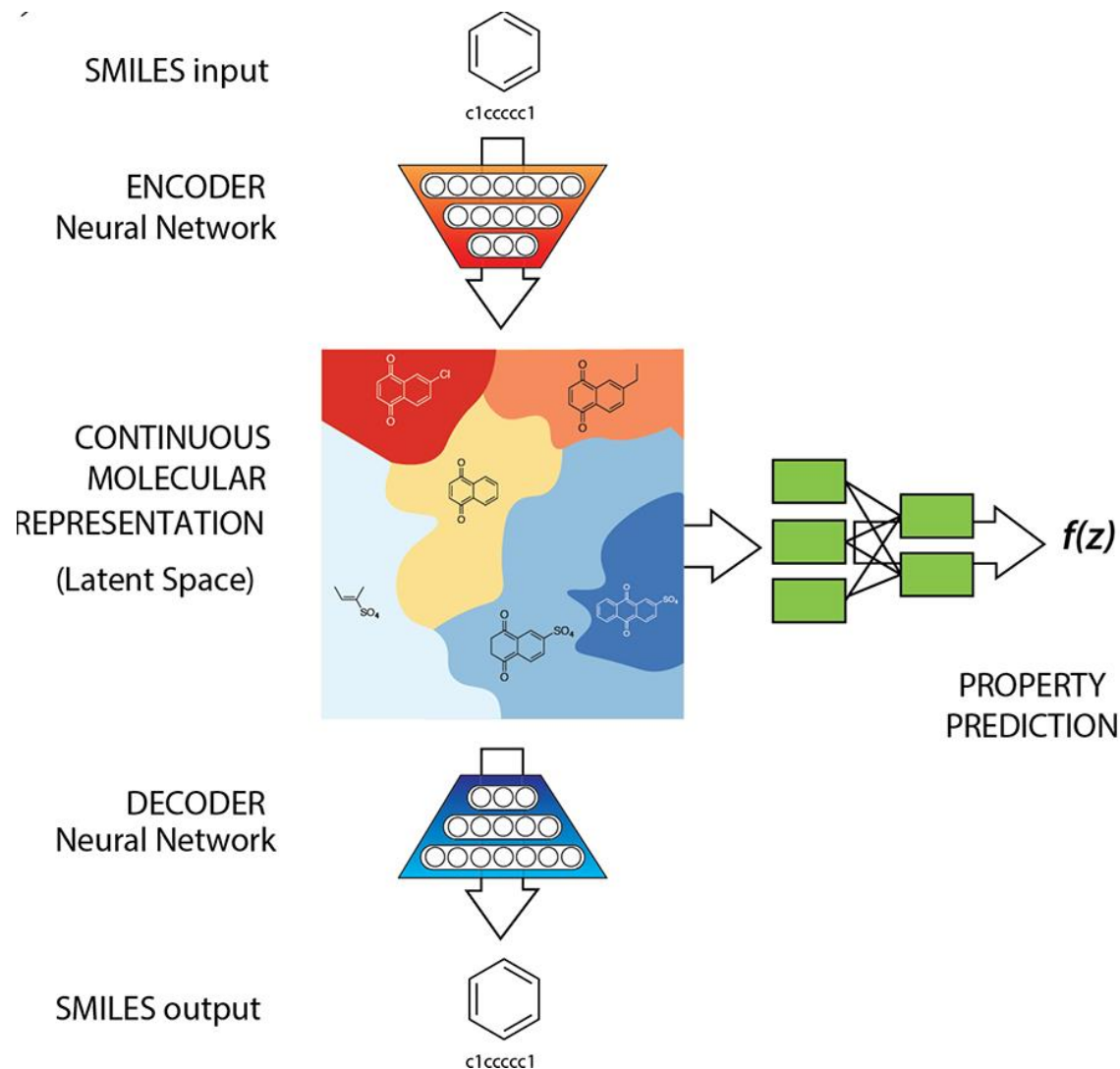


Cheng, Ser, Skreta, et al., Aspuru-Guzik, "Spiers Memorial Lecture: How to do impactful research in AI for chemistry and materials science," Faraday Discuss. 256, 10–60 (2025).

Direct vs Inverse design: example of redox flow batteries



Chemical VAE: turn molecules into a map to optimise



SECTION CLOSER · QUIZ

Quick quiz – True/False

1 In inverse design you specify the property you want and the model proposes a structure.

☒ True☐ False

2 A VAE gives a continuous latent space you can optimize in.

☒ True☐ False

3 The “third generation” of materials design is manual trial-and-error.

☐ True☒ False

Take-home messages - Design to a specification instead of stumbling onto it.

WHY IT MATTERS TO CLIMATE

- 1 We are in a ML-guided generation of materials design.
- 2 Inverse design = optimize in a learned latent space (continuous space of design).
- 3 There are two cross-cutting enablers: Foundation Models and Self-Driving Labs.

THE AI TOOLKIT

An AI toolkit

Six things ML actually does across the materials pipeline

1

Property prediction

PREDICT

Supervised models predict a property y (e.g. bandgap) from a structure x , aiding or replacing costly simulations.

2

Generative design

GENERATE

Unsupervised / generative models output new candidates x' , conditioned on target properties: inverse-design route.

3

Self-driving labs

SYNTHESIZE

Autonomous labs plan & run experiments: retrosynthesis, reaction prediction, active-learning optimization.

4

Aiding characterization

CHARACTERIZE

ML interprets measurements: crystal structure from TEM, coordination from XAS, symmetry from electron diffraction.

5

System management

DEPLOY

ML manages devices & grids: predicting battery life, forecasting loads, optimizing smart-grid control.

6

Accelerating theory

SIMULATE

ML makes intractable simulations feasible: cheaper potentials & functionals for larger scales.

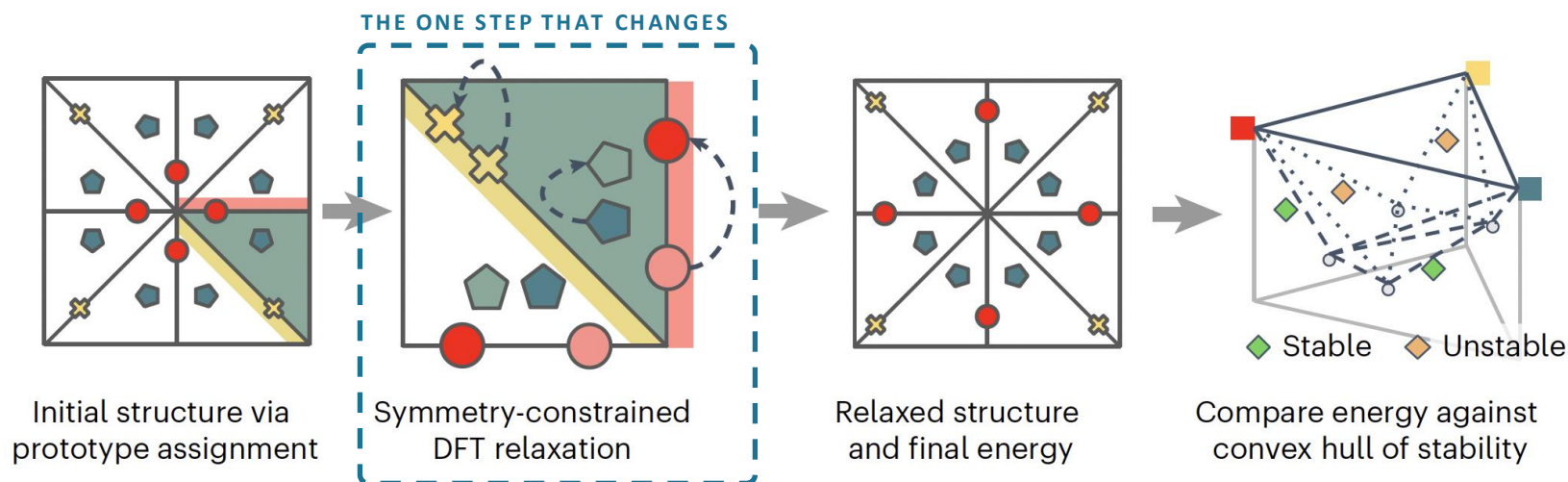
1

Property prediction

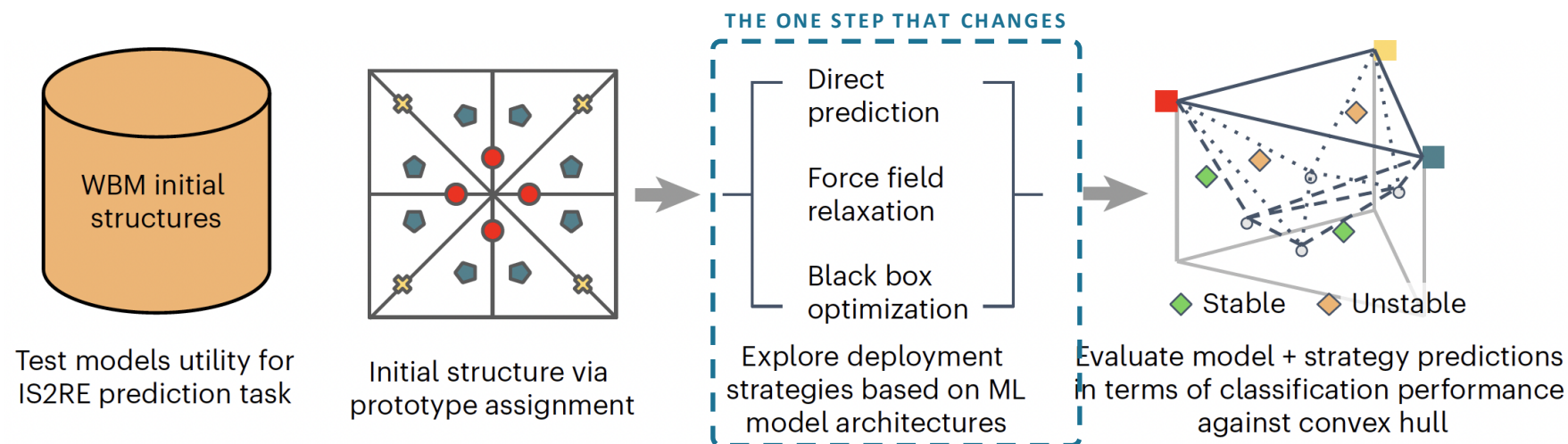
PREDICT

Predicting DFT properties with an ML surrogate

CONVENTIONAL



MATBENCH DISCOVERY

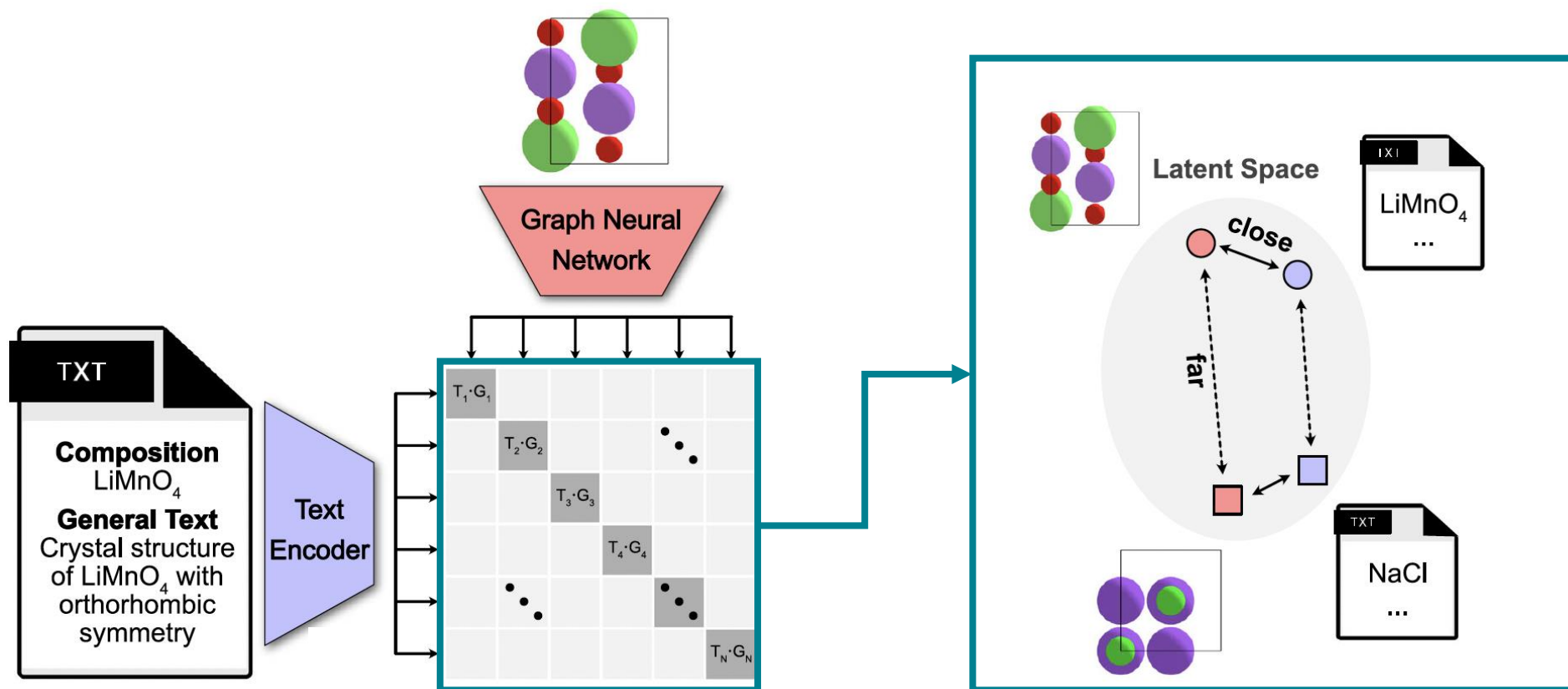
IS2RE = Initial
Structure to
Relaxed Energy

2

Generative / inverse design

GENERATE

Chemeleon: turning a text prompt into a crystal, in three steps

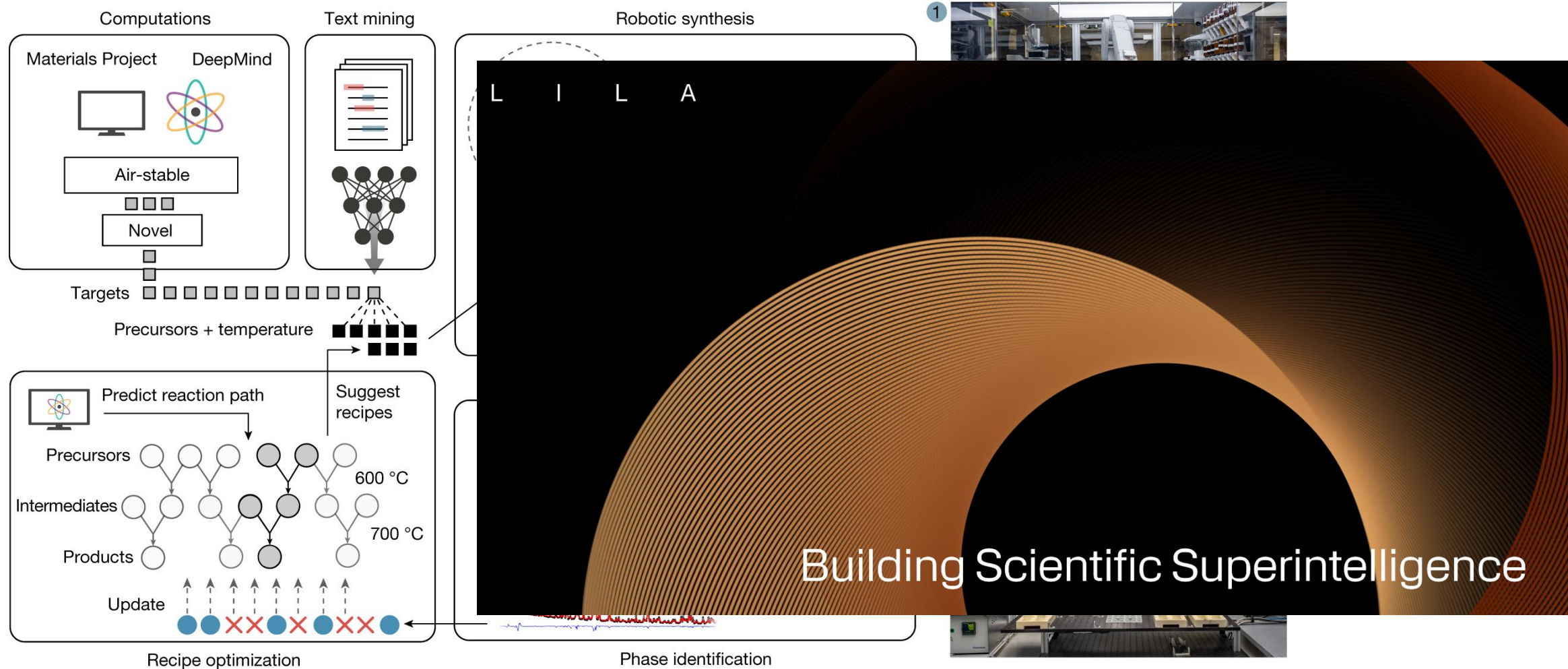


3

Self-driving labs

SYNTHESIZE

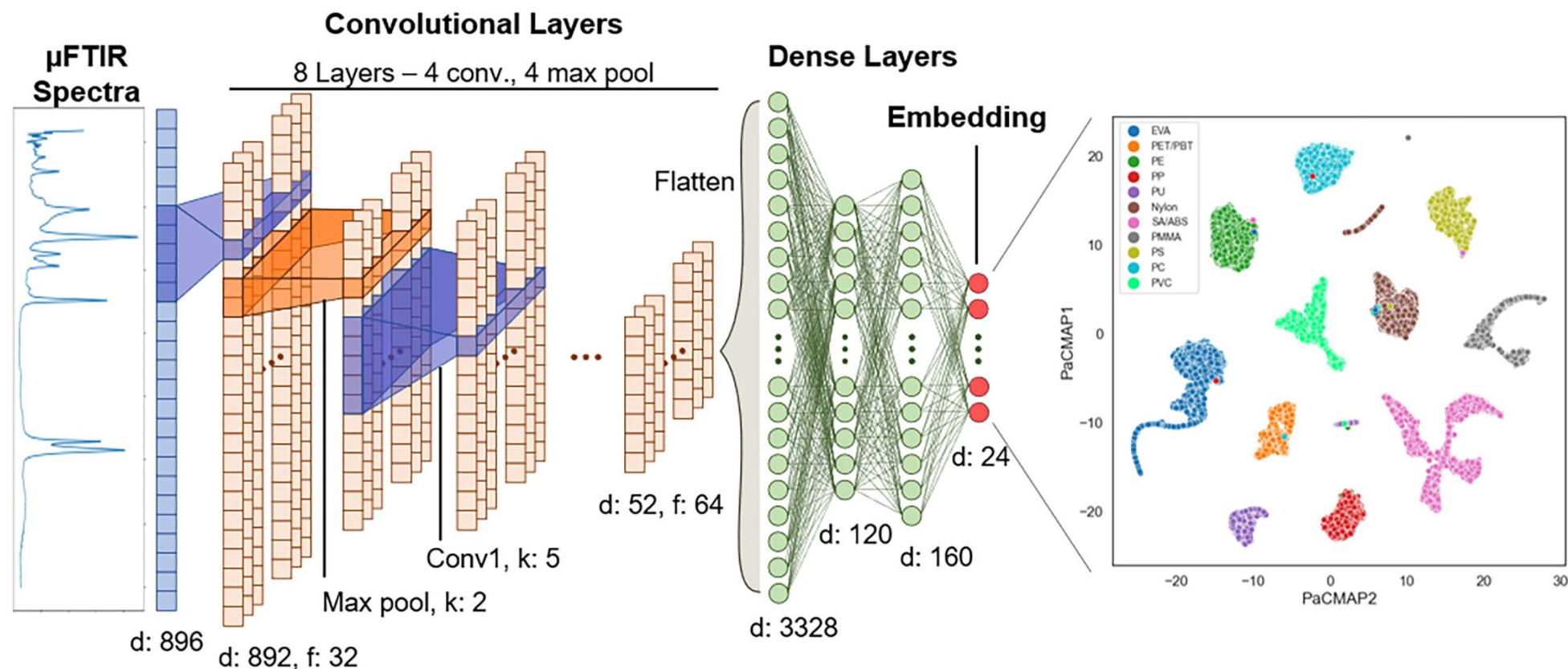
An autonomous laboratory: the A-Lab



4

Aiding characterization

CHARACTERIZATION

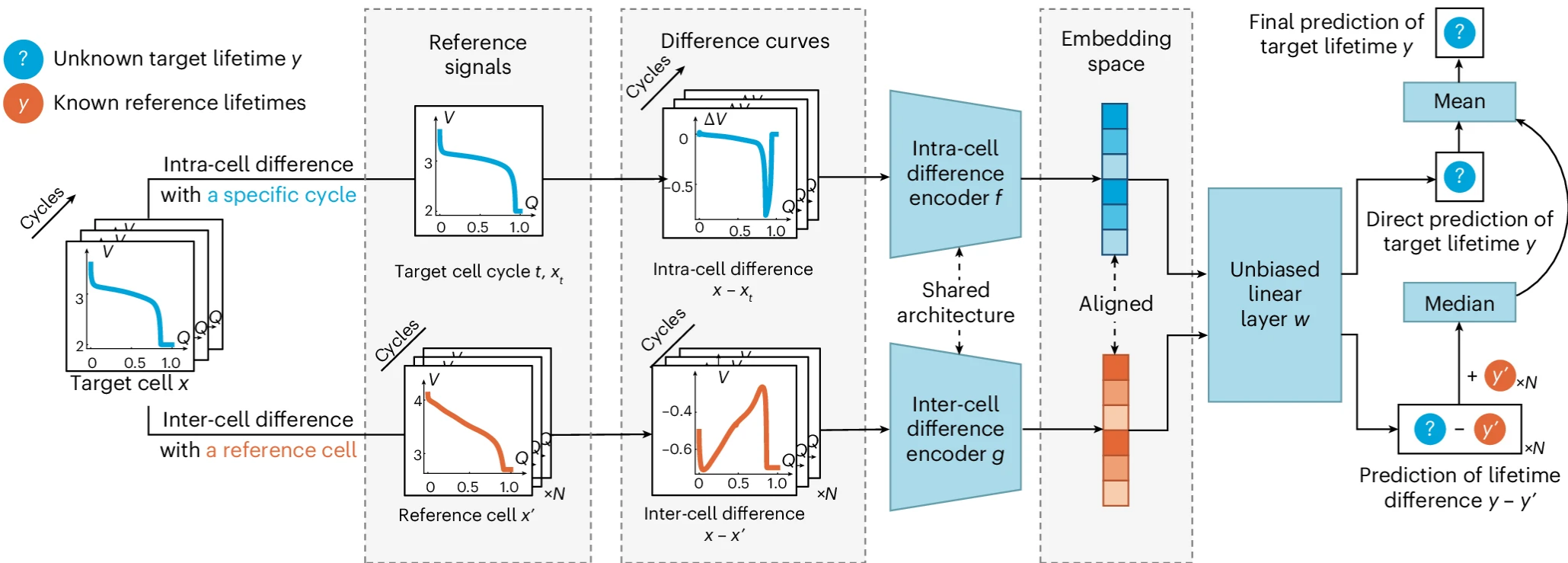
Similarity learning on μ FTIR spectral imaging of microplastics

5

System management

DEPLOY

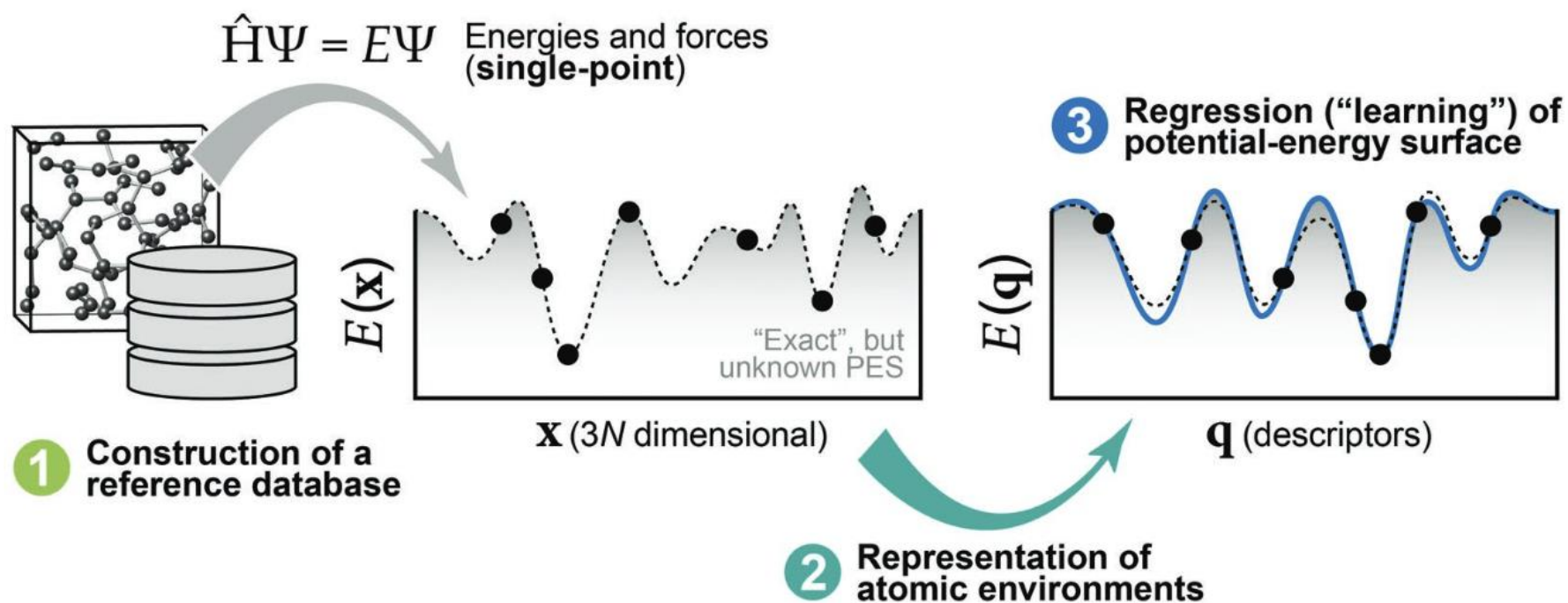
Predicting battery cycle life early



BatLiNet compares the target cell to a fully-cycled "reference" cell whose lifetime is already known

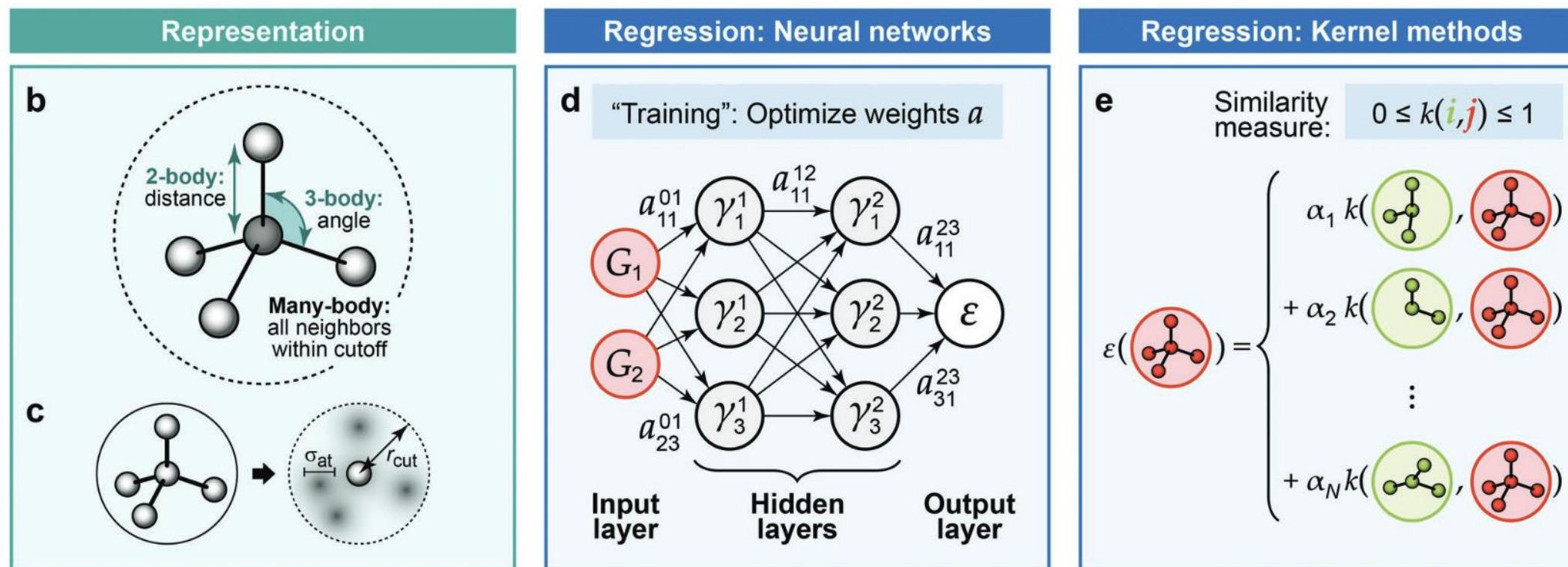
How a machine-learning interatomic potential works

One recipe underlies them all: local atomic environment $\rightarrow$ energy, learned from DFT



How a machine-learning interatomic potential works

One recipe underlies them all: local atomic environment $\rightarrow$ energy, learned from DFT



Atomic scale

Porosity, Atomic Structure, Carbon Hybridization, Structure Order



MLIP models at a glance

Representative models by family — key idea, typical use, and source

Model	Family	Key idea	Typical application	Reference
HDNNP	Descriptor + NN	symmetry functions → neural net	bulk solids, water	Behler & Parrinello 2007
GAP	Descriptor + kernel	SOAP + Gaussian process	Si, C, amorphous; data-efficient	Bartók et al. 2010
DeePMD	Learned descriptor + NN	end-to-end deep potential	large-scale MD (water, alloys)	Zhang et al. 2018
SNAP	Linear expansion	bispectrum components (linear)	metals (Ta, W)	Thompson et al. 2015
MTP	Linear expansion	moment tensors	alloys; active learning	Shapeev 2016
ACE	Body-ordered expansion	atomic cluster expansion	general; fast + accurate	Drautz 2019
SchNet	Invariant GNN	continuous-filter convolutions	molecules (MD17, QM9)	Schütt et al. 2018
NequIP	Equivariant GNN	E(3)-equivariant messages	data-efficient molecules + materials	Batzner et al. 2022
Allegro	Local equivariant GNN	strictly-local equivariant	very large-scale MD	Musaelian et al. 2023
MACE	Equivariant + high body-order	higher-order messages per layer	accurate & fast MD	Batatia et al. 2022
M3GNet	Universal GNN	MP-trained graph potential	crystal relaxation / screening	Chen & Ong 2022
CHGNet	Universal (charge-aware)	charge-informed GNN	Li-ion, oxides, magnetism	Deng et al. 2023
MACE-MP-0	Foundation (equivariant)	periodic-table foundation model	off-the-shelf MD across chemistries	Batatia et al. 2024

SOAP = smooth overlap of atomic positions · E(3) = 3-D rotations + translations + reflections · MP = Materials Project · GNN = graph neural network.

References

Machine-learning interatomic potentials, in roughly chronological order

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2. Bartók, A. P.; Payne, M. C.; Kondor, R.; Csányi, G. Gaussian Approximation Potentials (GAP). *Phys. Rev. Lett.* 104, 136403 (2010). 10.1103/PhysRevLett.104.136403
3. Thompson, A. P.; Swiler, L. P.; Trott, C. R.; Foiles, S. M.; Tucker, G. J. Spectral neighbor analysis method (SNAP). *J. Comput. Phys.* 285, 316–330 (2015). 10.1016/j.jcp.2014.12.018
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7. Drautz, R. Atomic cluster expansion (ACE). *Phys. Rev. B* 99, 014104 (2019). 10.1103/PhysRevB.99.014104
8. Batzner, S. et al. E(3)-equivariant GNNs for interatomic potentials (NequIP). *Nat. Commun.* 13, 2453 (2022). 10.1038/s41467-022-29939-5
9. Batatia, I.; Kovács, D. P.; Simm, G. N. C.; Ortner, C.; Csányi, G. MACE: Higher-Order Equivariant Message Passing. *NeurIPS* 35 (2022). arXiv:2206.07697
10. Chen, C.; Ong, S. P. A universal graph deep learning interatomic potential (M3GNet). *Nat. Comput. Sci.* 2, 718–728 (2022). 10.1038/s43588-022-00349-3
11. Musaelian, A. et al. Learning local equivariant representations (Allegro). *Nat. Commun.* 14, 579 (2023). 10.1038/s41467-023-36329-y
12. Deng, B. et al. CHGNet: charge-informed universal neural network potential. *Nat. Mach. Intell.* 5, 1031–1041 (2023). 10.1038/s42256-023-00716-3
13. Merchant, A. et al. Scaling deep learning for materials discovery (GNoME). *Nature* 624, 80–85 (2023). 10.1038/s41586-023-06735-9
14. Batatia, I. et al. A foundation model for atomistic materials chemistry (MACE-MP-0). arXiv:2401.00096 (2024).

SECTION CLOSER · QUIZ

Quick quiz – True/False

1 An ML interatomic potential aims for DFT-level accuracy at much lower cost.

☒ True☐ False

2 A model with 100% training accuracy is always the most trustworthy on new data.

☐ True☒ False

3 AI-generated candidate materials still need DFT / experimental validation.

☒ True☐ False

WHY IT MATTERS TO CLIMATE

One toolkit, every clean-energy material.

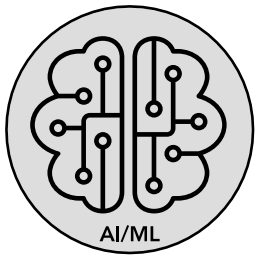
- ML potentials give near-DFT accuracy at a fraction of the energy and time: screen millions of candidates.
- Self-driving labs turn the best predictions into validated samples, fast.
- Validation keeps scarce lab effort off false positives.

Take-home messages

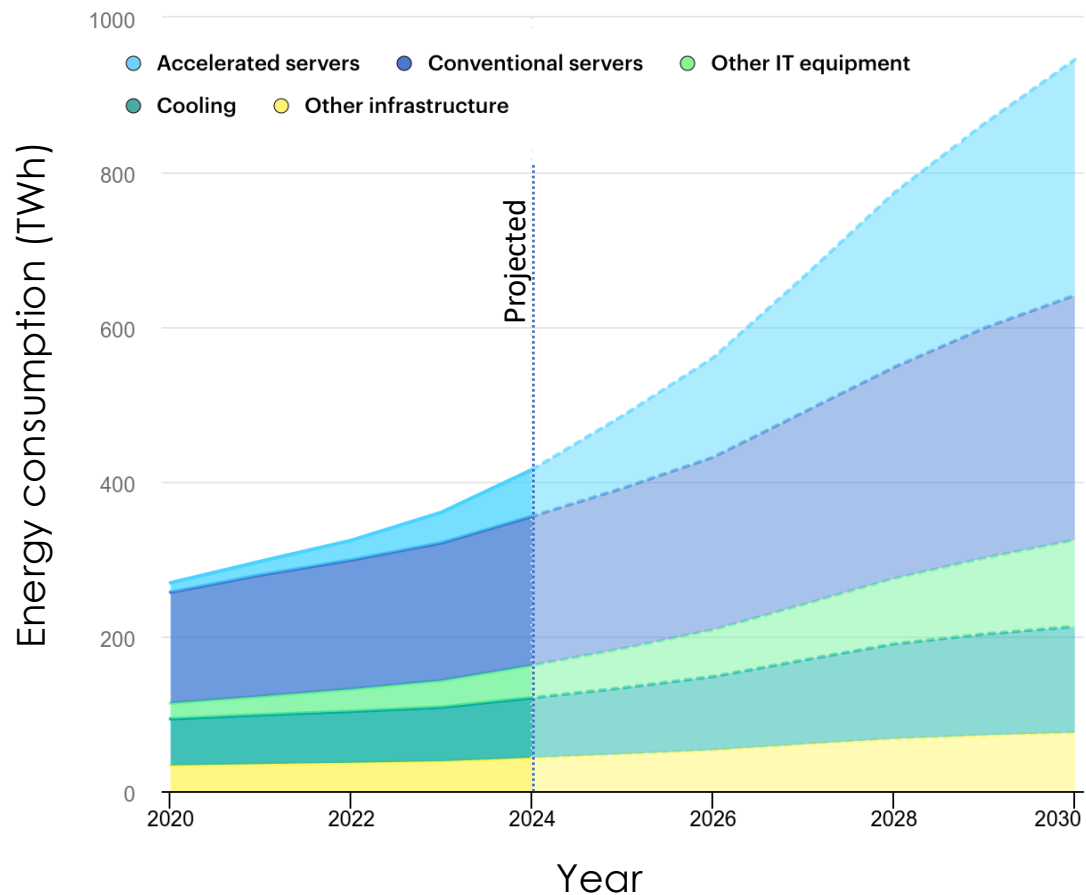
WHY IT MATTERS TO CLIMATE

- 1 Self-driving labs turn the best predictions into validated samples, fast.
- 2 MLIPs deliver DFT accuracy at MD speed: the “accelerate simulation” box made real.
- 3 Generation is easy; validation is hard

TALKING CHEMISTRY TO COMPUTERS

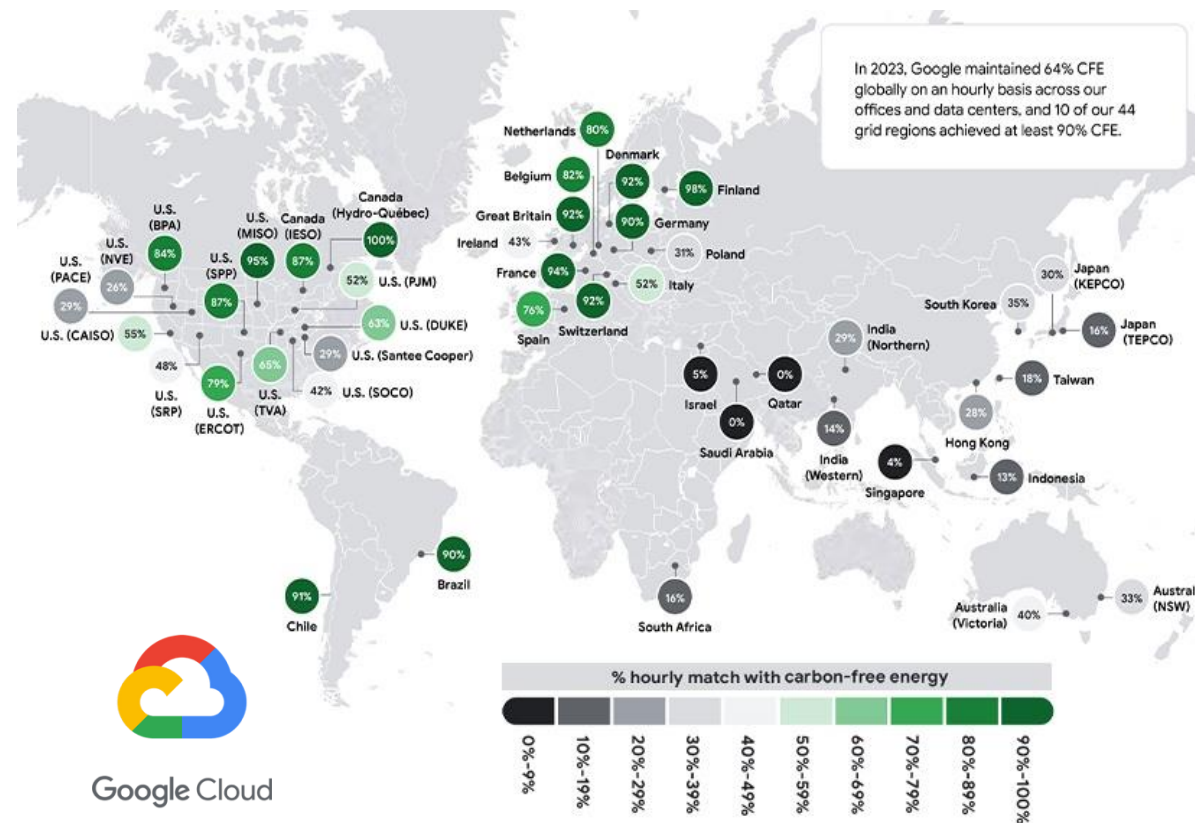


Increasing energy cost of AI and data centers



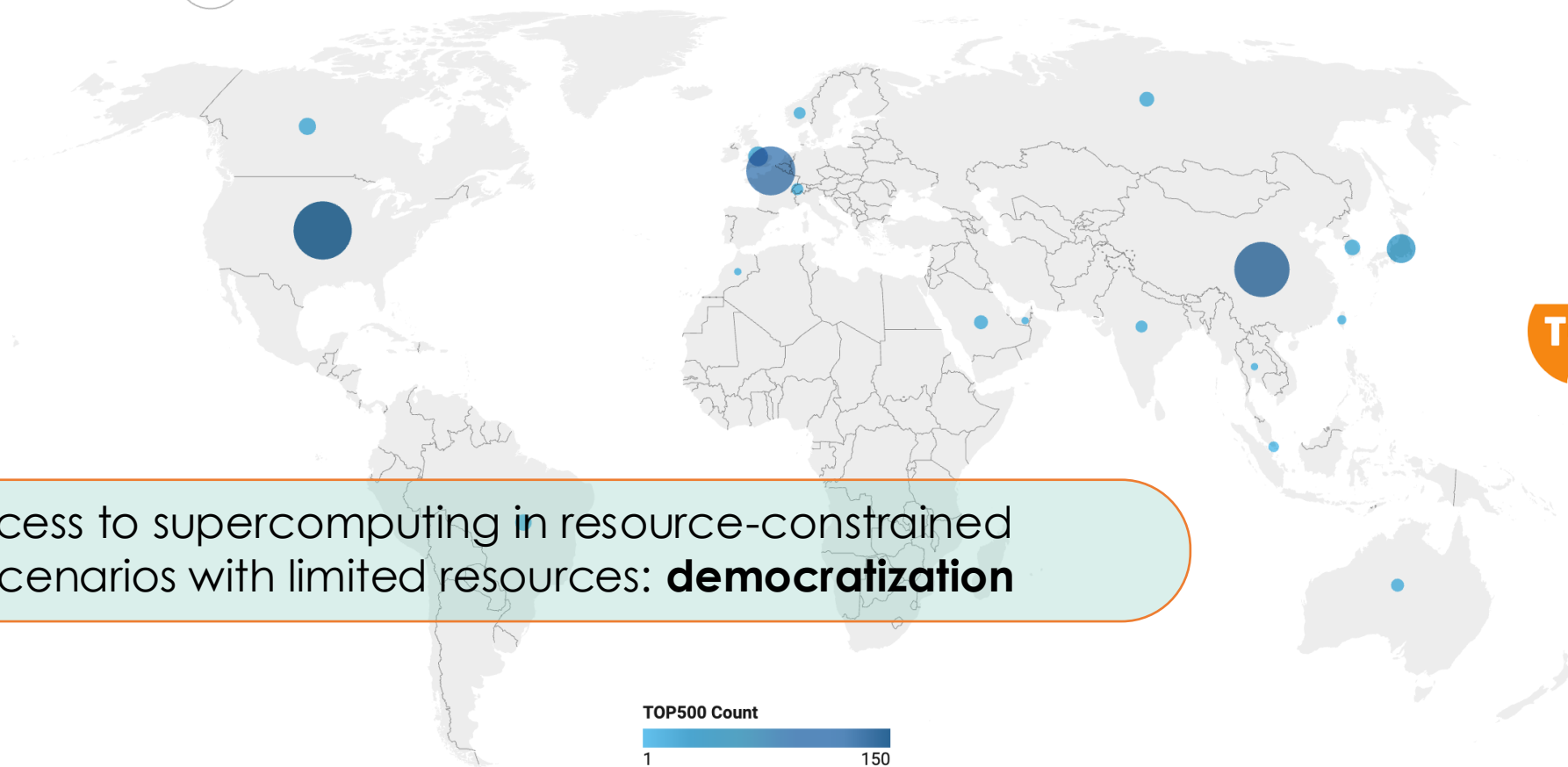
Source: International Energy Agency (IEA)

Need of greener and more democratized super computing

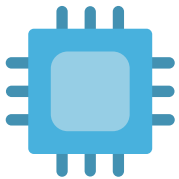


Source: Google Cloud

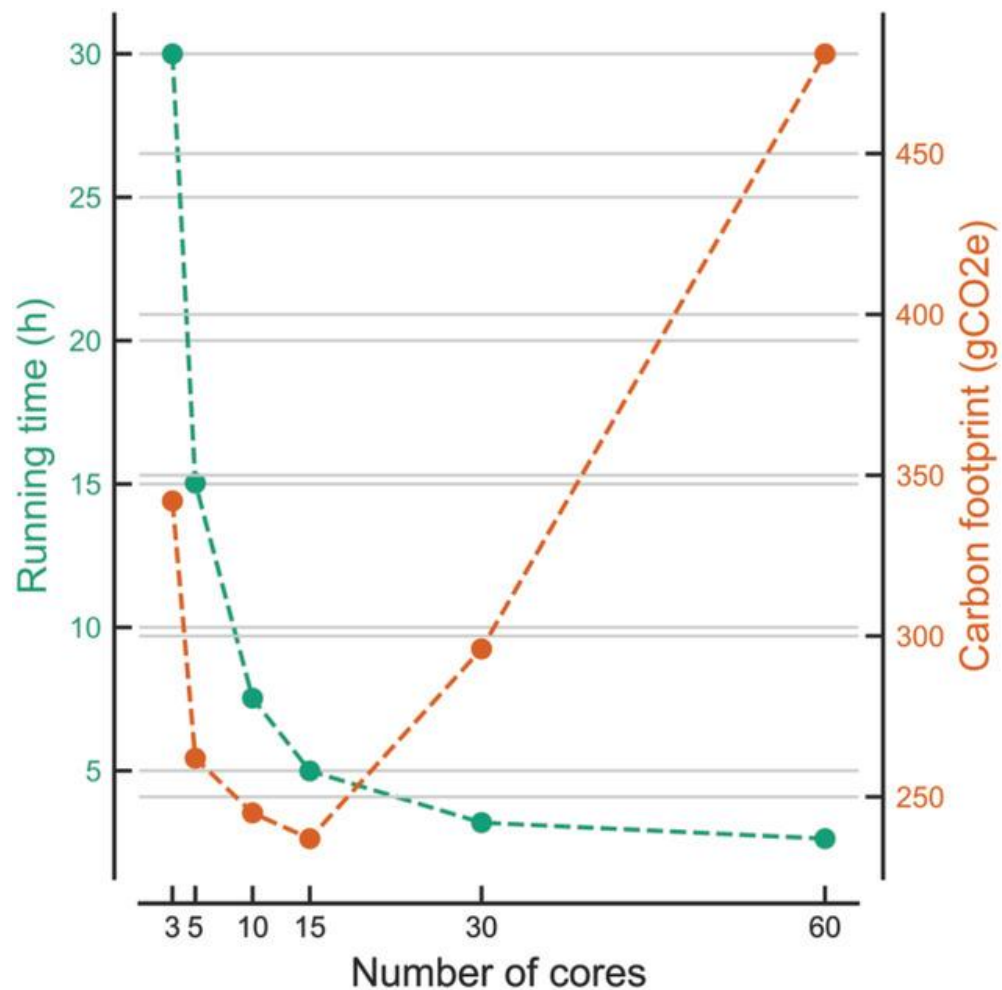
Where is supercomputing?



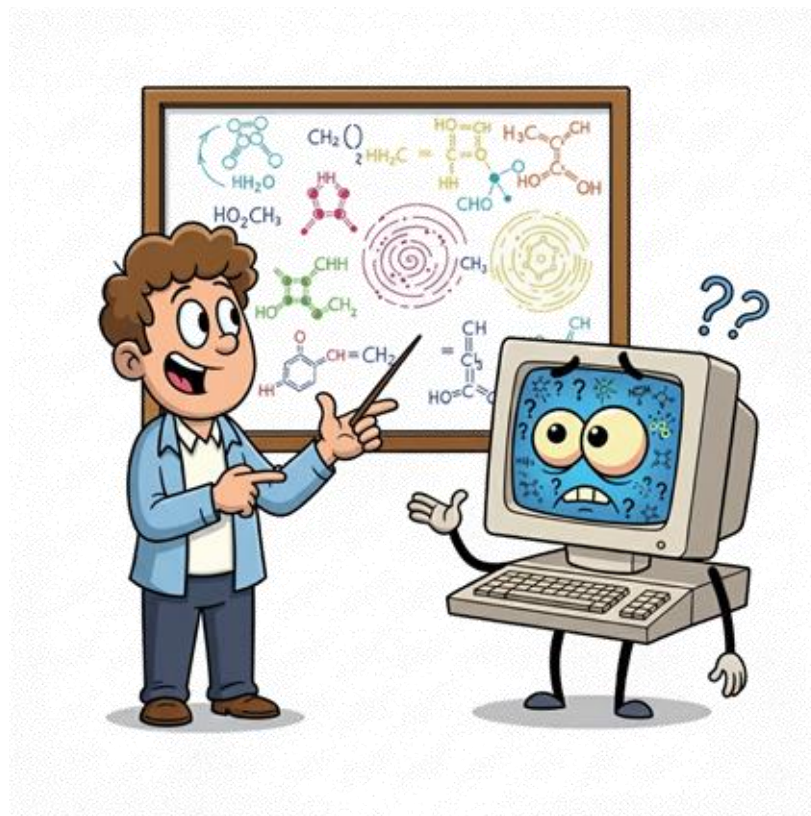
Improve access to supercomputing in resource-constrained devices or scenarios with limited resources: **democratization**



Green algorithms

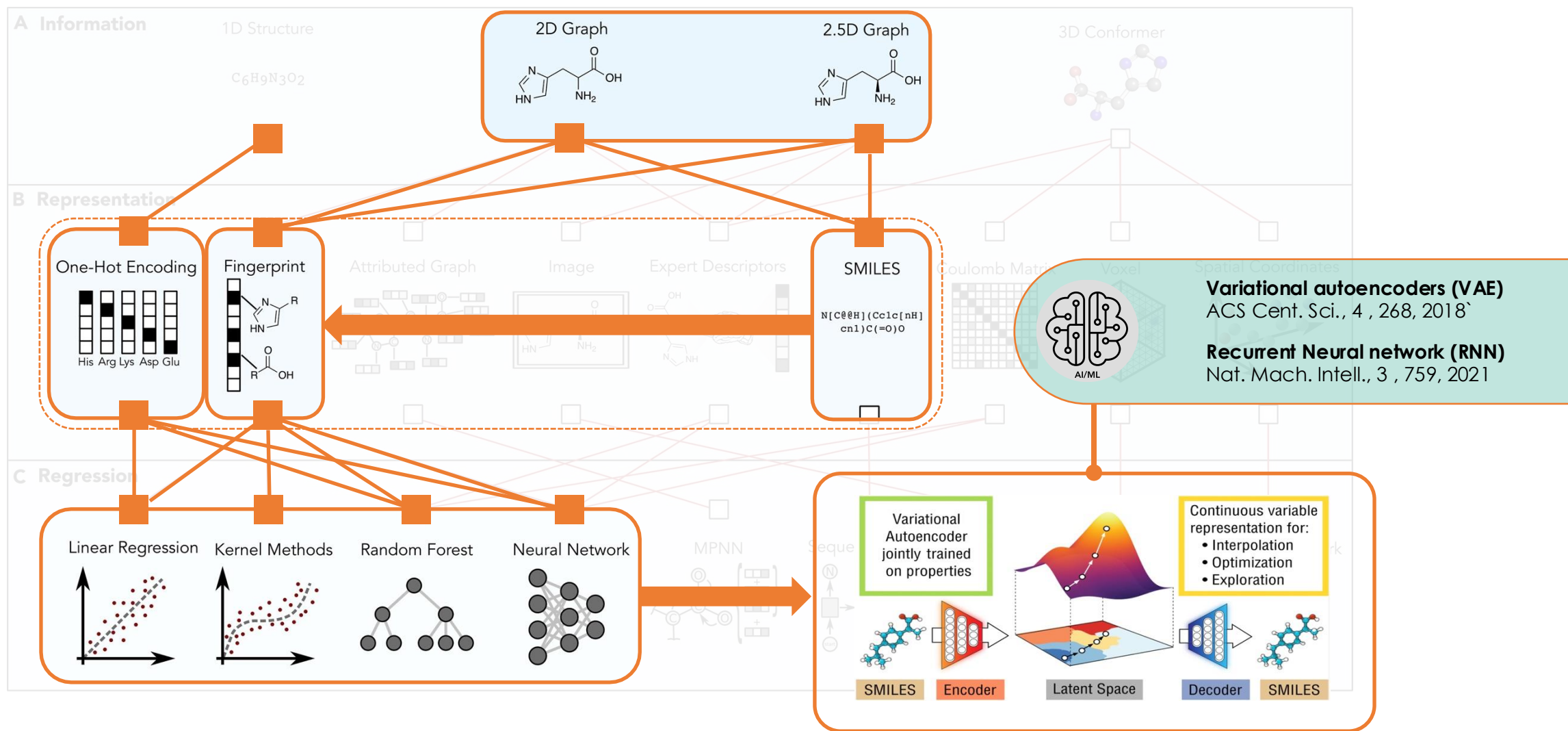


One step back: How do we talk chemistry to computers?



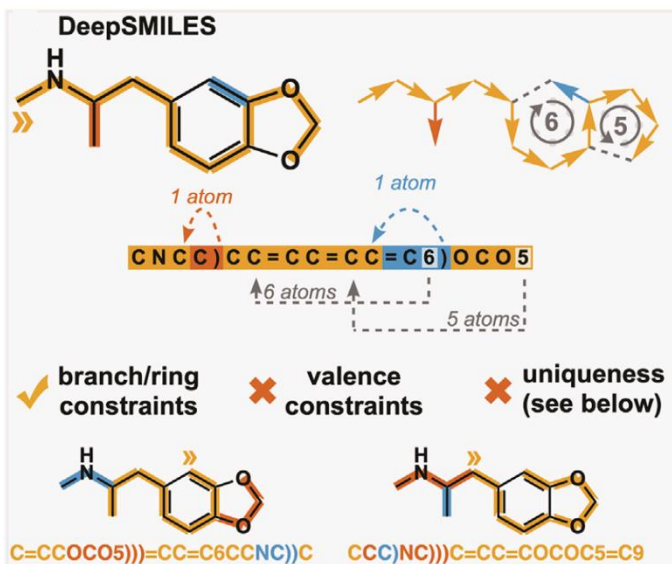
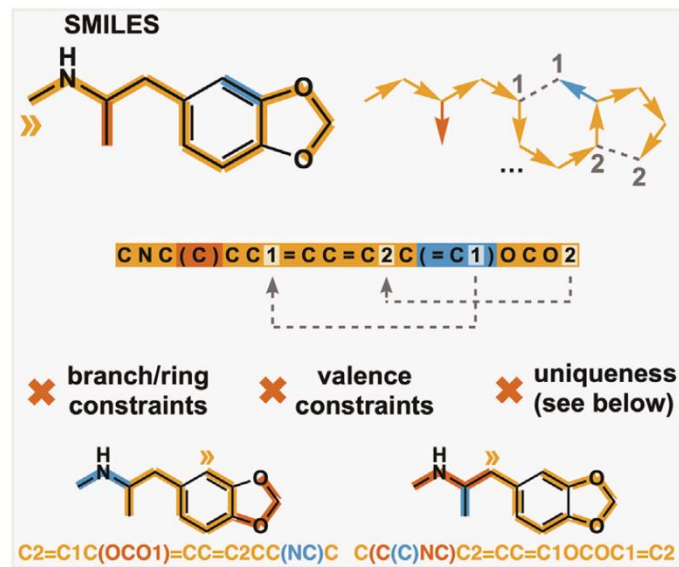
Machine learning for chemistry requires **featurizing molecules**

ML for chemistry requires featurizing molecules

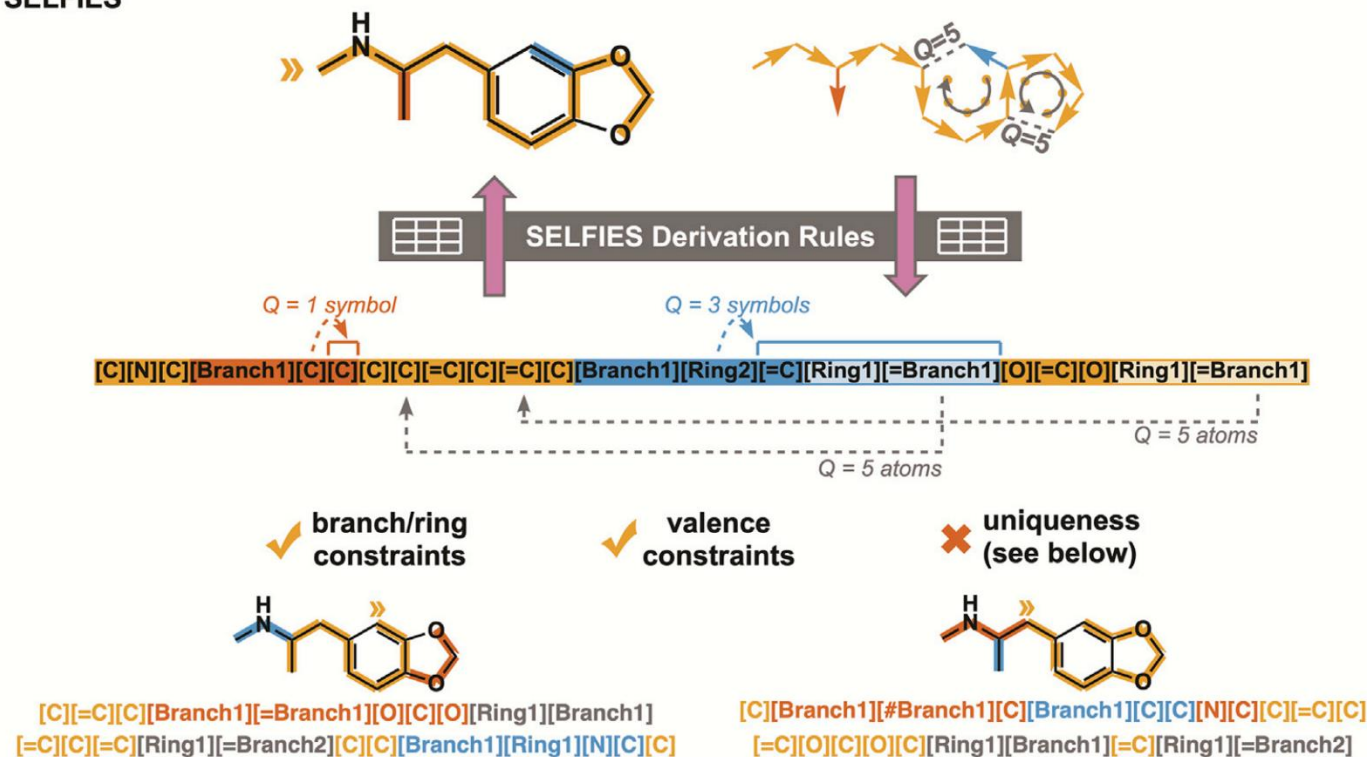




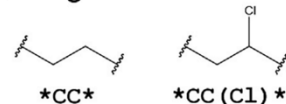
Machine-readable data encoding



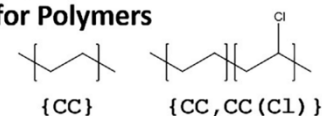
SELFIES



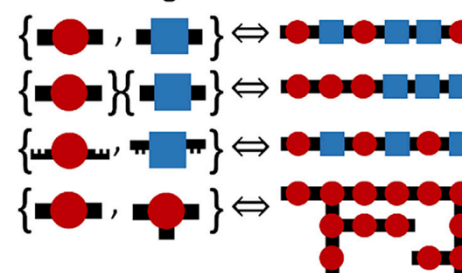
SMILES Representation for Organic Molecules

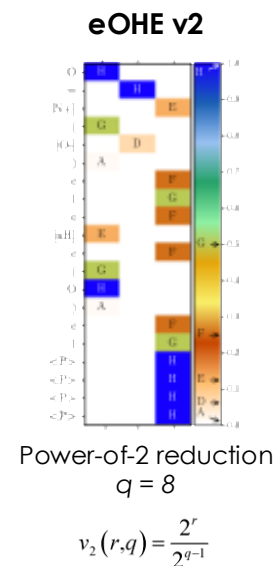
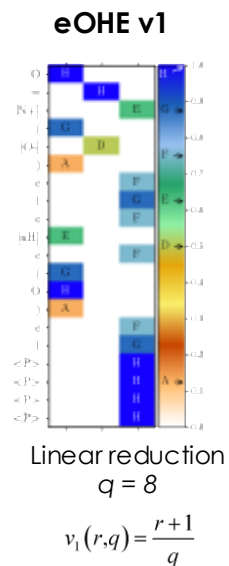
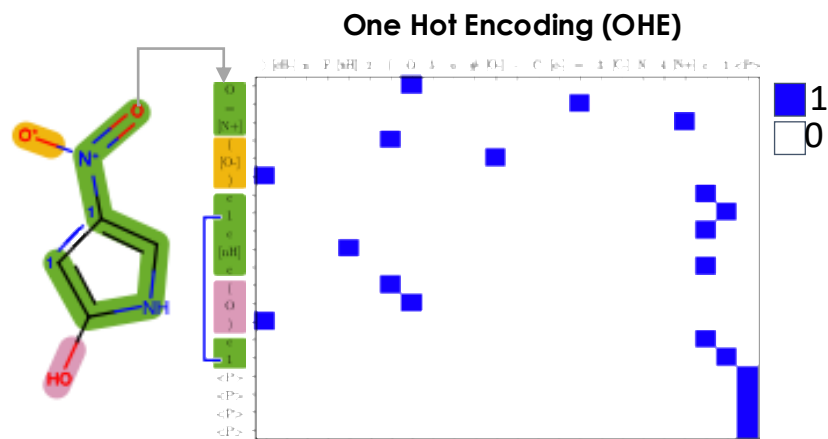


BigSMILES Representation for Polymers

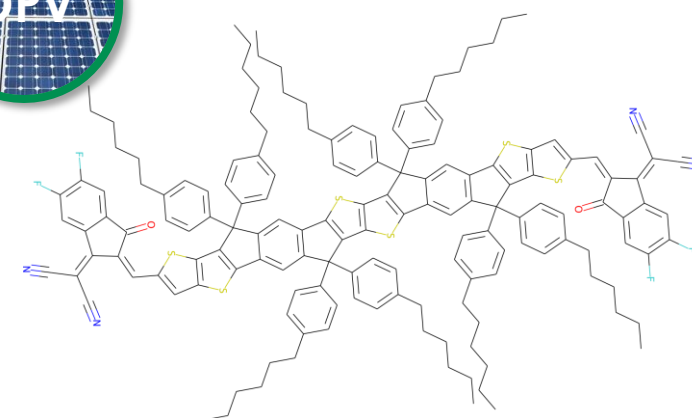
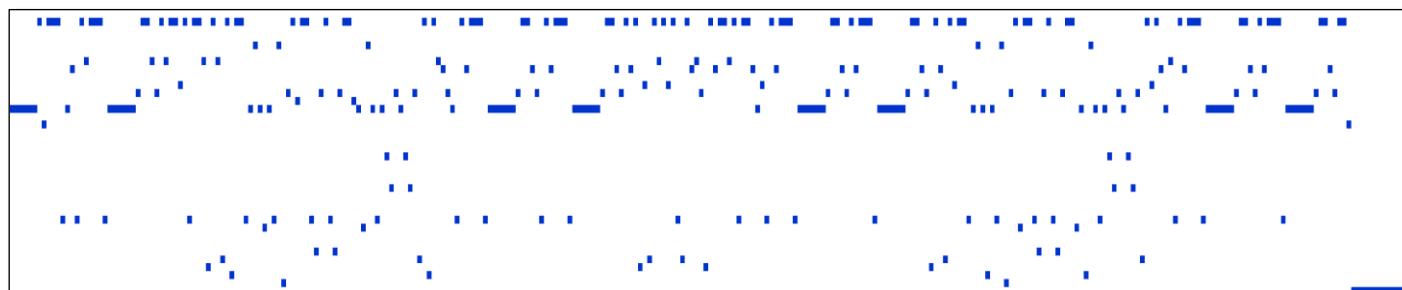


BigSMILES Supports a Wide Range of Structures

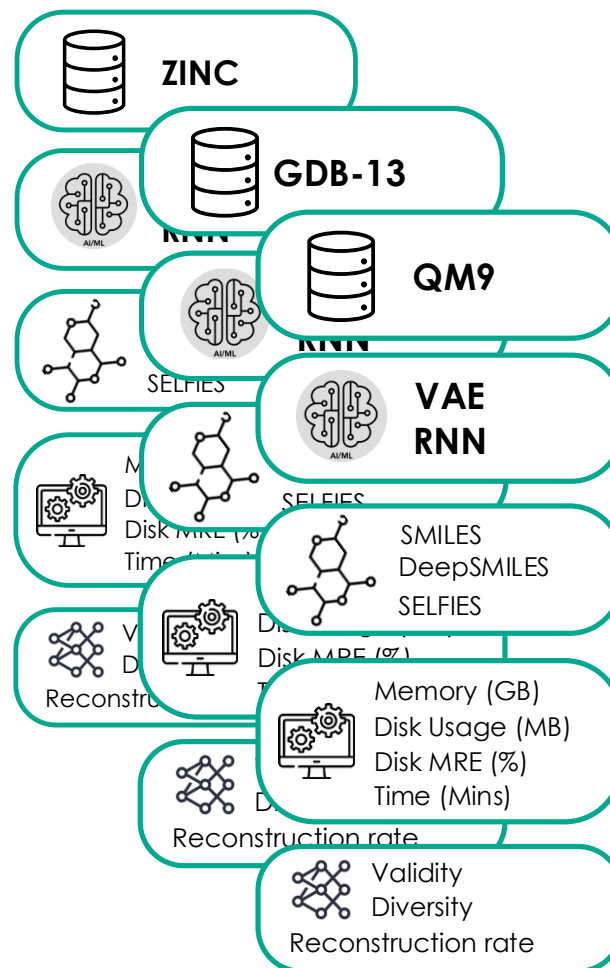
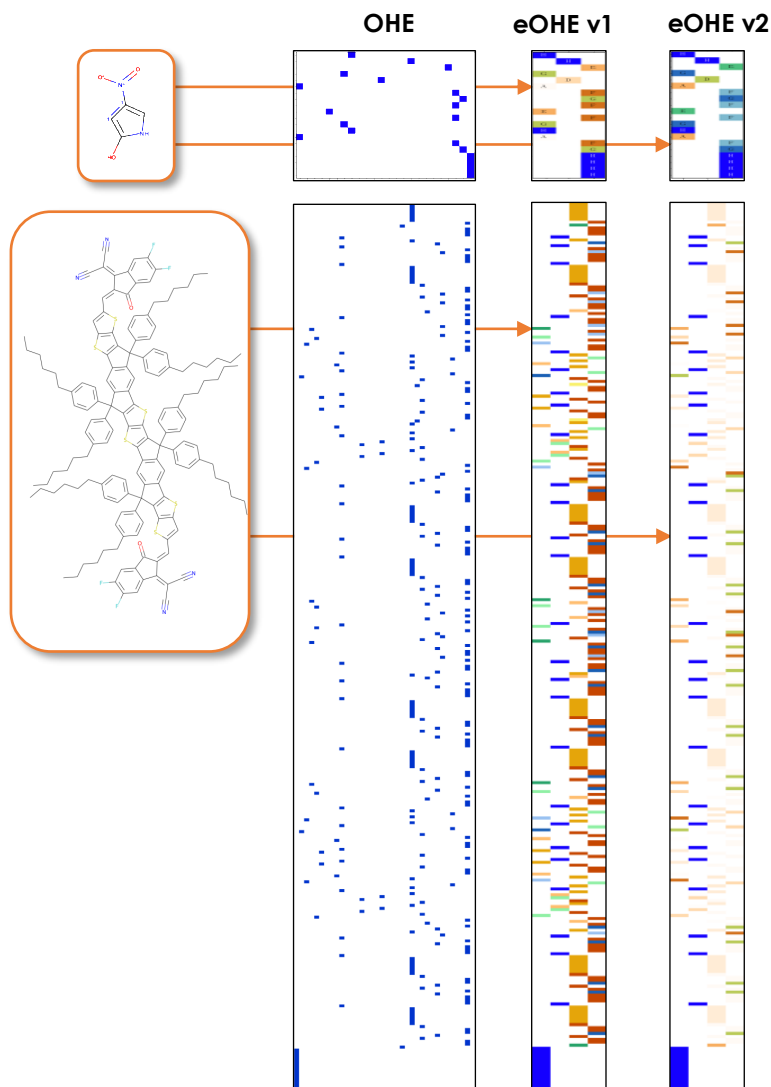




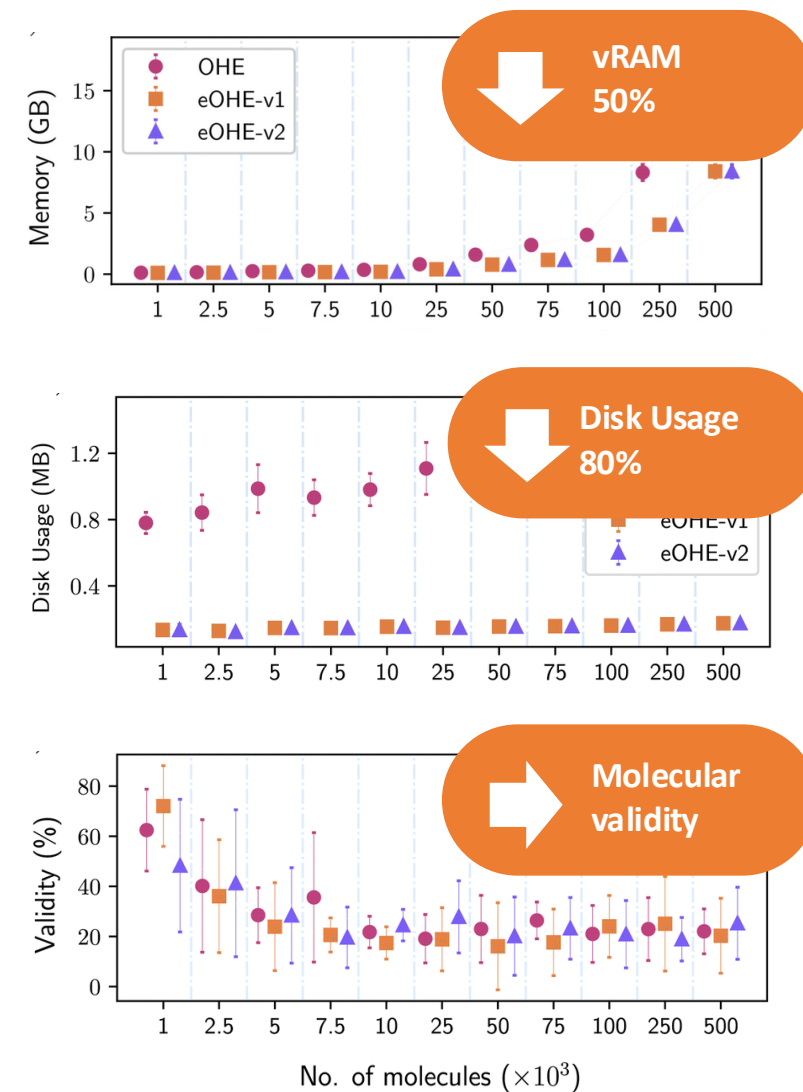
Organic photovoltaics Donor/Acceptors

CCCCCc1ccc(C2(c3ccc(CCCCCC)cc3)c3cc4c(cc3-c3sc5ccc(=C=C6C(=O)C7c(F)(F)C7C6=C([C#N])[C#N])sc5C32)C2cccc(CCCCCC)cc2)(c2ccc(CCCCCC)cc2)-c2c-4sc3c4c(CCCCCC)c2cc3c(cc2C2cccc(CCCCCC)cc2)c2ccc(CCCCCC)cc2)-c2sc4ccc(C=C5C(=O)C6c(F)(F)C6C5=C([C#N])[C#N])sc4c2C32)(c2ccc(CCCCCC)cc2)c2ccc(CCCCCC)cc2cc1

More efficient machine-readable codification

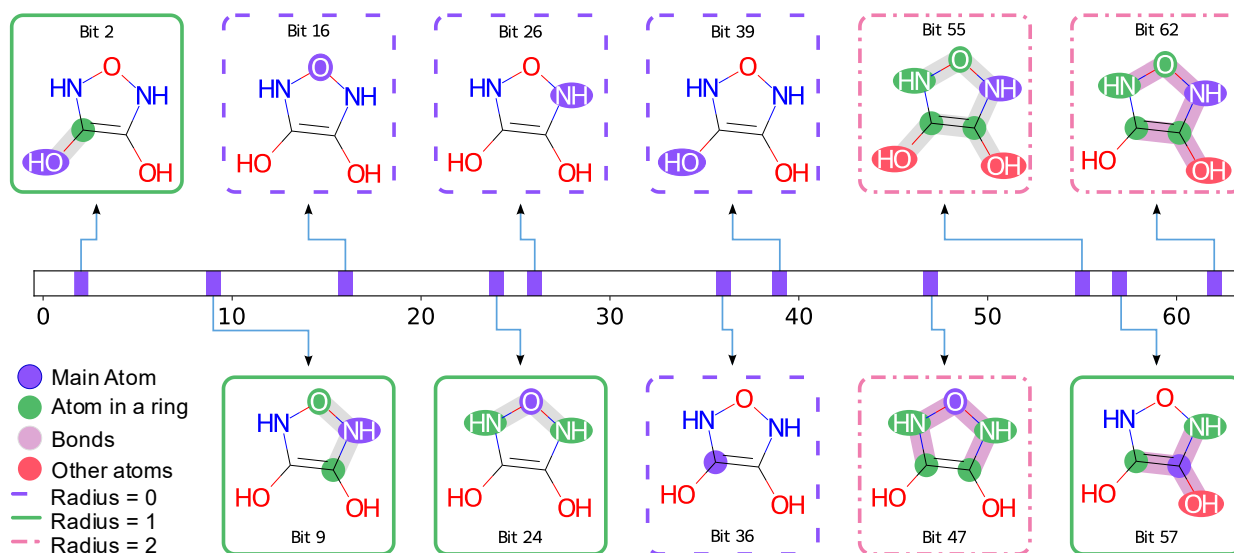


Performance in variational autoencoders



embedded Morgan Fingerprints (eMP)

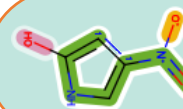
A Morgan fingerprint is a fixed-size binary vector that encodes the presence (1's) or absence (0's) of substructures in a molecule.



QM9: 134000+ molecules
NFA: 50000 molecules
RedDB: 15000 molecules



**Random Forest (RF),
Multi-layer Perceptron (MLP),
K-Neighbors Regressor (KNR),
Gradient Booster Regressor (GBR)
Deep Neural Network (DNN)**

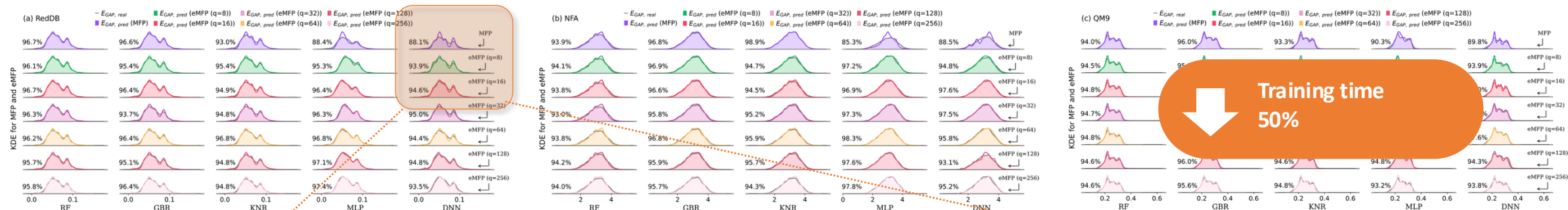


MFP
eMFP (q=8, 16, 32, 64, 128, 256)



**150 trials for 15750
hyperparameters
optimisation**

Energy gap with embedded Morgan Fingerprints (eMP)

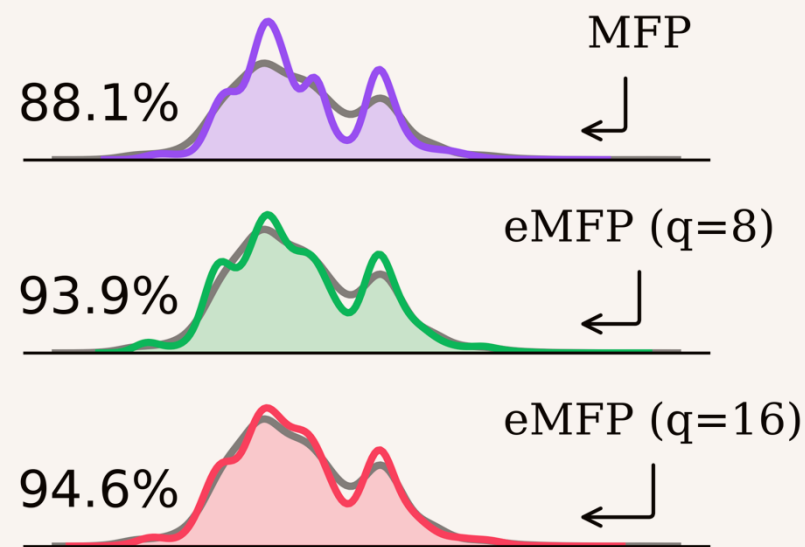


RedDB Data Base

— $E_{GAP, real}$ ■ $E_{GAP, pred} (MFP)$

■ $E_{GAP, pred} (eMFP (q=8))$

■ $E_{GAP, pred} (eMFP (q=16))$



SECTION CLOSER · QUIZ (multiple choice)

Multiple choice

Q1 Which representation guarantees a valid molecule for every string?

A) SMILES

B) SELFIES

C) InChI

D) None of these

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A) Accuracy only

C) Publication count

B) Memory / disk / time at matched accuracy

D) Nothing

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A) It doesn't

C) Compute is free

B) Sustainability & democratization

D) Only for gaming

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Take-home messages - Representation is a sustainability decision.

WHY IT MATTERS TO CLIMATE

- 1 Featurization is a first-class design decision, not a preprocessing afterthought.
- 2 eOHE / eMFP cut memory, disk and time at matched accuracy
efficiency = greener, more democratized compute.
- 3 Green, accessible computing is part of responsible acceleration.

LLM-GUIDED MATERIALS DEVELOPMENT

LLM-guided materials development: a new layer on the stack

Beyond predicting numbers, models that read, reason, plan and use tools

The shift: narrow ML maps one structure to one property. A large language model adds a natural-language interface, broad prior knowledge, step-by-step planning, and the ability to call external tools, **turning a predictor into a research collaborator**.

1

Language interface

Ask, adapt with a few examples, strong in the low-data limit where classical ML struggles.

2

World knowledge

Pretrained on the scientific literature, so it brings chemical priors and context, not just curve-fitting.

3

Planning & reasoning

Decompose a goal ("find a stable Na-ion cathode") into ordered, checkable steps.

4

Tool use

Call simulators, databases and robots, grounding answers in real computation instead of guessing.

...but be careful. LLMs are fluent yet unreliable. They hallucinate facts, mis-cite, and can be confidently wrong. Everything depends on **grounding in tools and validation/**

What LLMs actually do in materials development

Six capabilities — each grounded in a concrete, published example

1 Read the literature

Turn thousands of papers into structured, queryable databases (compositions, MOFs, dopants).

e.g. [Dagdelen et al. 2024](#)

2 Predict properties

Answer chemistry questions in natural language; competitive with ML, strong in the low-data limit.

e.g. [Jablonka et al. 2024](#)

3 Generate & inverse-design

Write molecules and crystals as text; property-directed generation.

e.g. [Gruver et al. 2024](#) · [Chemeleon](#)

4 Plan synthesis

Propose synthetic routes and recipes; retrosynthesis and procedure design.

e.g. [ChemCrow 2024](#)

5 Reason with tools

Augment the model with expert tools (RDKit, calculators, search) so it stops guessing.

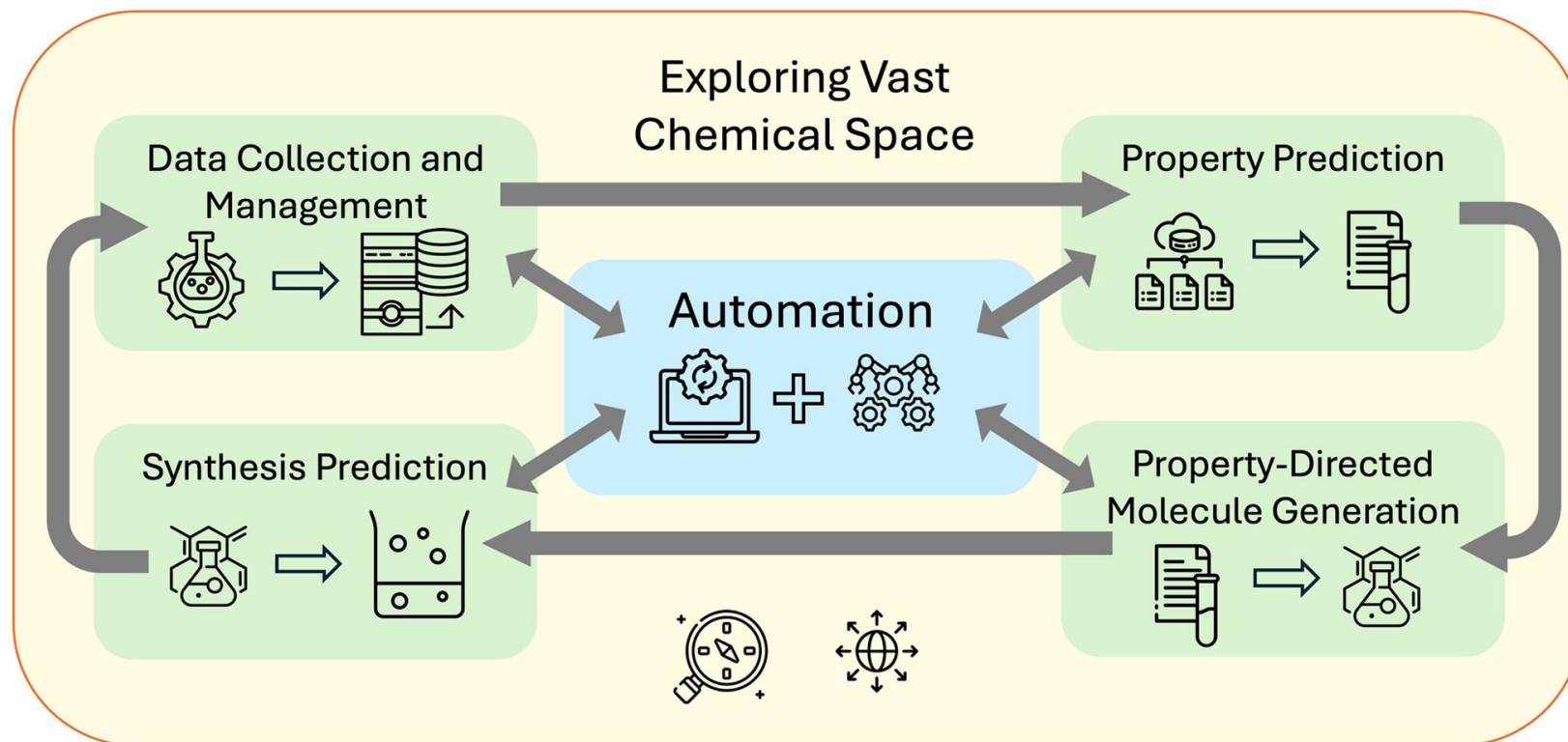
e.g. [ChemCrow — 18 tools](#)

6 Run experiments autonomously

Plan-act-observe loops that drive instruments and robots to test hypotheses.

e.g. [Coscientist 2023](#)

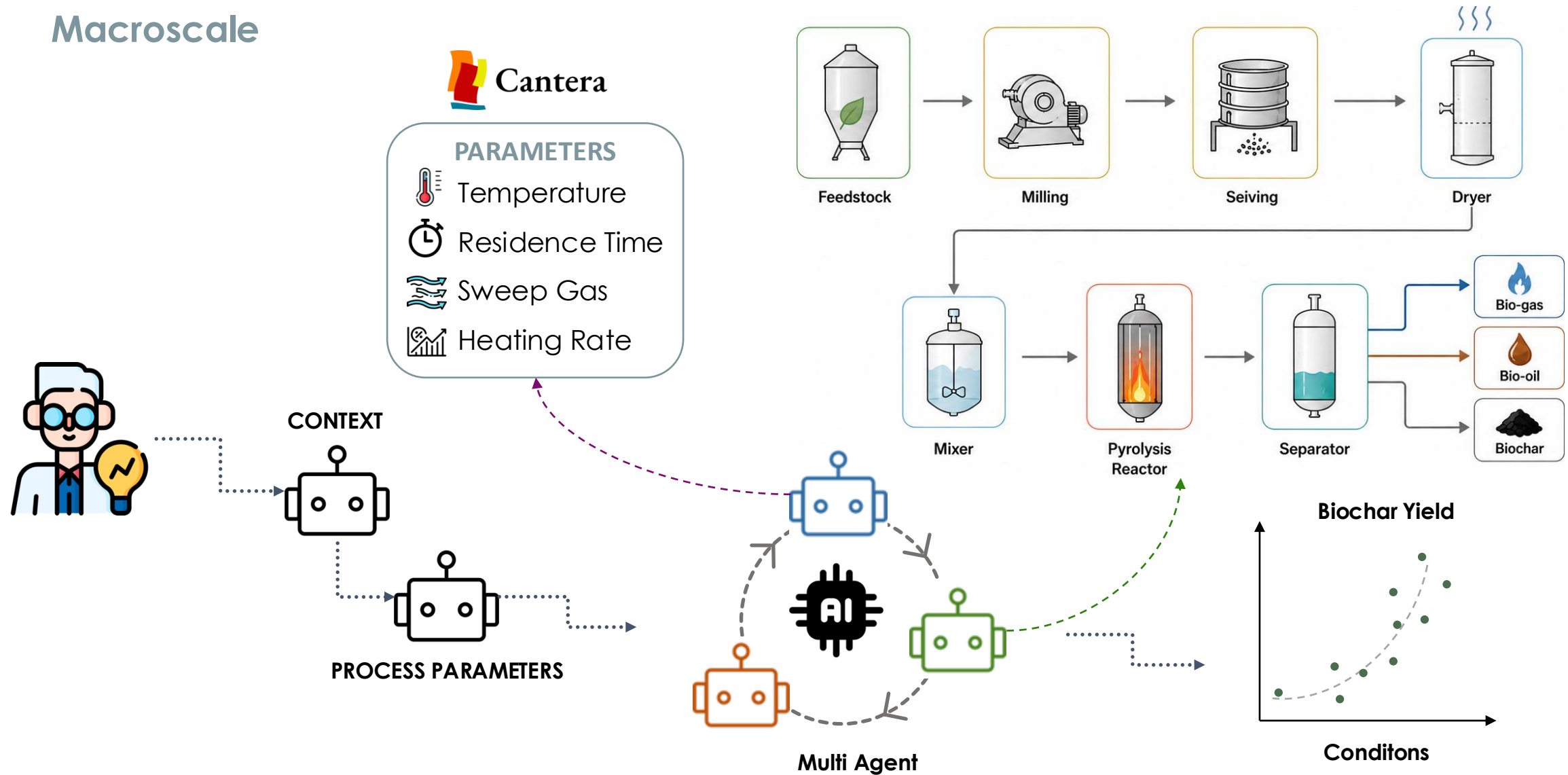
What can LLMs and agents actually do in chemistry?



Modeling biochar – pyrolysis process engineering

LLM-GUIDED DEVELOPMENT

Macroscale



Landmark LLM systems for chemistry & materials

Representative systems. What they do, the approach, and the source

System	What it does	LLM / approach	Reference
Coscientist	designs, plans & runs experiments (e.g. Pd cross-couplings)	GPT-4 + web/docs/code + robotics	Boiko et al., Nature 2023
ChemCrow	organic synthesis, drug & materials design; ran real syntheses	GPT-4 + 18 expert tools	Bran et al., Nat. Mach. Intell. 2024
Predictive-chem LLM	property prediction & classification from text; low-data	fine-tuned GPT-3; in-context	Jablonka et al., Nat. Mach. Intell. 2024
Struct. extraction	reads papers → structured records (dopants, MOFs, compositions)	fine-tuned GPT-3 / Llama-2	Dagdelen et al., Nat. Commun. 2024
Crystal-text LLM	generates stable inorganic crystals written as text	fine-tuned LLaMA-2 70B	Gruver et al., ICLR 2024
Chemeleon	text-guided crystal-structure generation	contrastive text-crystal + diffusion	Park et al., Nat. Commun. 2025

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SECTION CLOSER · QUIZ (multiple choice)

Multiple choice

Q1 An LLM “agent” (vs assistant) is one that...

- A) Only answers questions
- B) Plans and acts via tools in a loop
- C) Never uses tools
- D) Runs only offline

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Q3 LLMs are especially useful when...

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- B) Data are scarce (low-data limit)
- C) Never
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 - C) Never
 - D) Only for images

Take-home messages - Unlock the knowledge trapped in text.

WHY IT MATTERS TO CLIMATE

- 1 LLMs add a language + reasoning + tool-use layer over narrow ML.
- 2 Mine decades of literature into usable data; plan syntheses; orchestrate tools.
- 3 Keep a human accountable and log every action. Provenance and oversight are what make autonomy trustworthy (and reproducible).

RESPONSIBLE AI



Responsible AI for materials & chemistry

Powerful tools bring responsibilities. The same acceleration cuts both ways

1 Integrity & reproducibility

Data leakage and benchmark-chasing inflate results.
Report enough that others can reproduce them.

2 Trustworthy predictions

Report uncertainty, respect the applicability domain, and validate downstream (DFT → experiment).

3 Data quality & bias

Datasets skew to well-studied, successful systems.
Garbage in, garbage out

4 Openness & FAIR

Share data, code and model weights
Findable, Accessible, Interoperable, Reusable (FAIR).

5 Safety & dual use

Powerful design tools can be inverted for harm.
Access control and human oversight.

6 Sustainability & compute

Training and inference cost energy.
Right-size models and reuse pretrained ones.

Challenges

- Implicit beliefs, and biases from Western contexts may be normalized in AI, making it less globally responsible.
- A need to employ a global lens in core AI assumptions: training data, model performance, and **explainability** of systems
- Ask the difficult questions on positive and negative impacts early on, rather than introducing repairs and post-hoc fixes.

Actions

- Be Robust to Multilingualism and Diverse Literacies
- Personalize Experiences with the Shared Identity Model
- Explainability and Transparency Should Make Sense for New Technology Users
- Proactively Address Fairness in Data Skews

Anthropogenic biases in chemical reaction data hinder exploratory inorganic synthesis

[Xiwen Jia](#), [Allyson Lynch](#), [Yuheng Huang](#), [Matthew Danielson](#), [Immaculate Lang'at](#), [Alexander Milder](#), [Aaron E. Ruby](#), [Hao Wang](#), [Sorelle A. Friedler](#) ✉, [Alexander J. Norquist](#) ✉ & [Joshua Schrier](#) ✉

[Nature](#) **573**, 251–255 (2019) | [Cite this article](#)

Machine-learning-assisted materials discovery using failed experiments

[Paul Raccuglia](#), [Katherine C. Elbert](#), [Philip D. F. Adler](#), [Casey Falk](#), [Malia B. Wenny](#), [Aurelio Mollo](#), [Matthias Zeller](#), [Sorelle A. Friedler](#) ✉, [Joshua Schrier](#) ✉ & [Alexander J. Norquist](#) ✉

[Nature](#) **533**, 73–76 (2016) | [Cite this article](#)

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Responsible and trustworthy AI for science

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SECTION CLOSER · QUIZ (true / false)

True or false?

1 Published chemistry datasets are biased partly because failed experiments are rarely reported.

☒ True☐ False

2 A drug-design model can be inverted to design toxins.

☒ True☐ False

3 FAIR stands for Findable, Accessible, Interoperable, Reusable.

☒ True☐ False

Take-home messages - Acceleration only helps if it is trustworthy and shared.

WHY IT MATTERS TO CLIMATE

- 1 Data are biased: publication bias, human bias, Western-context defaults.
- 2 Powerful generative/agentic tools are inherently dual-use.
- 3 Reproducibility, openness (FAIR) and compute cost are part of doing this well.

EXPLAINABLE AI

Explainable AI: from prediction to understanding

In science we usually want the “why”, not just the number

The gap: a model can predict a property accurately and still teach you nothing. Explainability asks what the model learned, whether it is for the right reasons, and what new science it implies.

1 Understanding

Turn a fitted model into design rules and mechanisms. Recover the physics/chemistry behind the data.

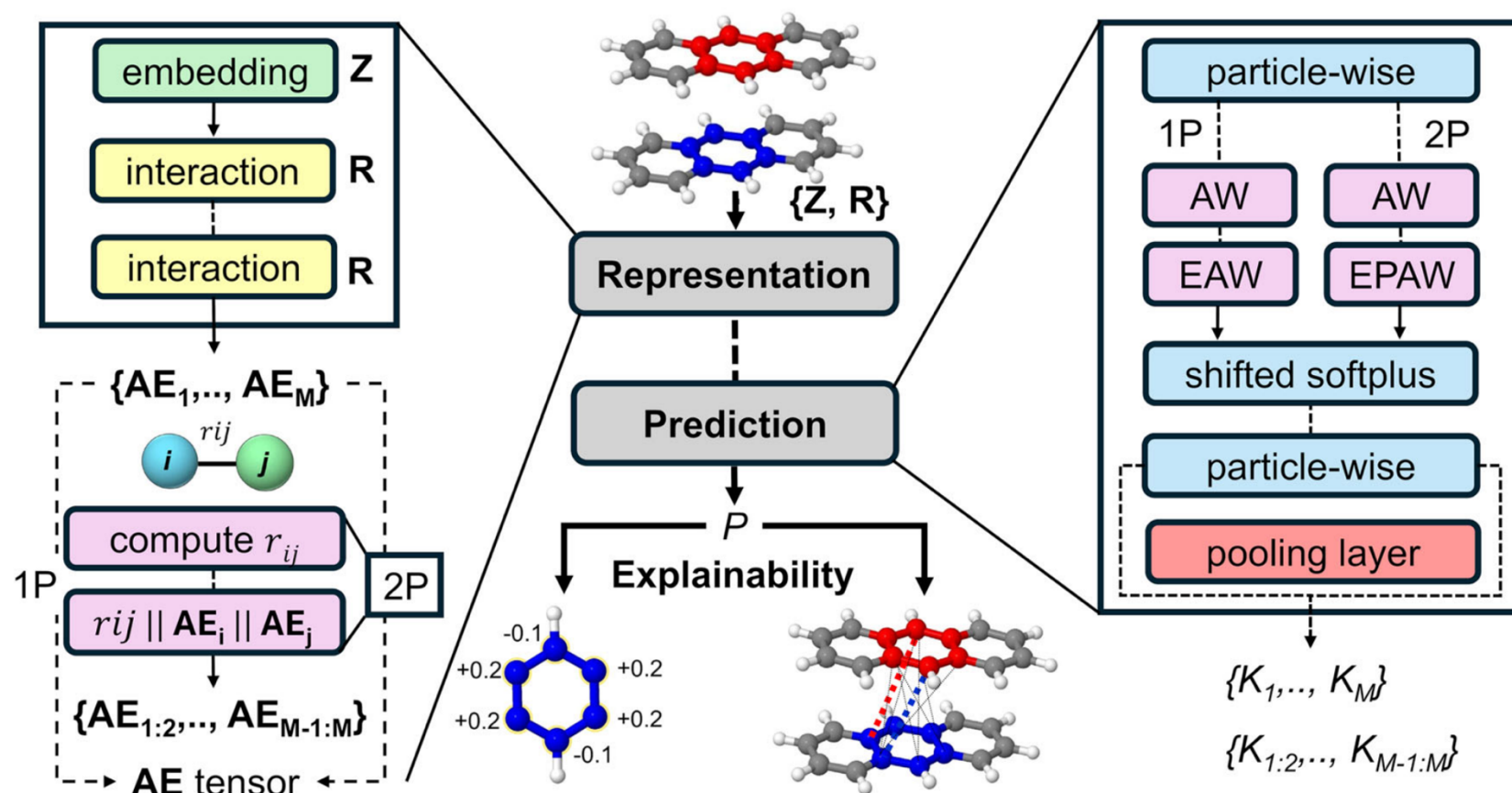
2 Trust & validation

Check the model predicts for the right reasons, not false correlations or dataset artefacts.

3 Discovery

Interpretation surfaces new structure–property relationships and testable hypotheses.

Explainable Chemical AI: SchNet4AIM



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Explainable & interpretable AI for science and materials

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Take-home messages - Turn fast predictions into design rules we can trust.

WHY IT MATTERS TO CLIMATE

- 1 In science we want the why, not just the number.
- 2 Interpretable methods expose models that are right for the wrong reasons.
- 3 Treat every explanation as a hypothesis to verify against physics and experiment.

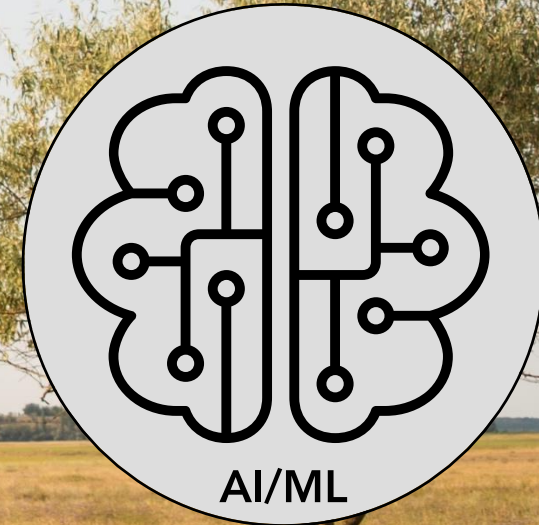
FINAL REFLECTIONS

Why AI cannot do good science without humans

With the arrival of 'AI scientists', it's as well to remember that human wisdom, empathy and sheer messiness are as much part of progress as are process and efficiency.



Design for circularity
Design for performance
as well as performance



Online repositories — open-source toolkit for AI-accelerated materials science

A shareable starting kit. Our group's tools are highlighted; verify links are current before distributing.

Area	Tool	What it's for	Repository
Data & tools	pymatgen	Materials analysis, I/O & Materials Project API	github.com/materialsproject/pymatgen
Data & tools	matminer	Ready-made descriptors / featurization for ML	github.com/hackingmaterials/matminer
Data & tools	RDKit	Cheminformatics: molecules, fingerprints, descriptors	github.com/rdkit/rdkit
Data & tools	DeepChem	ML-for-chemistry toolkit: datasets, models, benchmarks	github.com/deepchem/deepchem
Data & tools	ASE	Atomistic Simulation Environment: build / run / analyze	gitlab.com/ase/ase
Representations	SELFIES	Always-valid molecular string representation	github.com/aspuru-guzik-group/selfies
Representations	eOHE	Embedded one-hot encodings (efficient inputs)	github.com/MMLabCodes/eOHE
Representations	eMFP	Embedded Morgan fingerprints (compact, faster)	github.com/MMLabCodes/eMFP
Generative	chemical_vae	Continuous latent-space molecular design	github.com/aspuru-guzik-group/chemical_vae
Generative	MatterGen	Diffusion generative model for inorganic crystals	github.com/microsoft/mattergen
ML potentials	MACE	ML interatomic potentials (+ MACE-MP-0 foundation)	github.com/ACEsuit/mace
ML potentials	CHGNet	Universal charge-informed neural network potential	github.com/CederGroupHub/chgnet
ML potentials	MatGL (M3GNet)	Universal graph potentials + property models	github.com/materialsvirtuallab/matgl
Autonomy	Ax / BoTorch	Bayesian optimization for self-driving labs	github.com/facebook/Ax
LLM agents	ChemCrow	LLM chemistry agent wired to expert tools	github.com/ur-whitelab/chemcrow-public
Explainability	SHAP	Feature-attribution explanations for any model	github.com/shap/shap
Explainability	SISSO	Interpretable descriptor / formula discovery	github.com/rouyang2017/SISSO
Green computing	CodeCarbon	Track the carbon cost of your compute	github.com/mlco2/codecarbon

Highlighted rows = our group's repositories (github.com/MMLabCodes). List is representative, not exhaustive.

Further reading

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Thank you

[M_M]^{LAB}